

$$\frac{\partial n_a}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} \tilde{\Gamma}_{ax} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \right) = S_a^n, a = 0, 1, \dots, n_s-1$$

$$\tilde{\Gamma}_{ax} = \begin{cases} \left(b_x V_{\parallel a} + V_{ax}^{(E)} + V_{ax}^{(AN)} + V_a^{corr-dpc} + 0 \cdot \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis||)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{en_a} \right) n_a - \\ - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x} - \tilde{\Gamma}_{ax}^{(PSch)}, \quad z_{n_a} = 1 \ \& \ z_a = 1 \\ \left(b_x V_{\parallel a} + V_{ax}^{(E)} + V_{ax}^{(AN)} + V_a^{corr-dpc} \right) n_a - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x} - \tilde{\Gamma}_{ax}^{(PSch)} \end{cases}$$

$$\tilde{\Gamma}_{ay} = \begin{cases} \left(V_{ay}^{(E)} + V_{ay}^{(AN)} + \frac{j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + \tilde{j}_y^{(visq)}}{en_a} \right) n_a - \\ - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \frac{1}{h_y} \frac{\partial p_a}{\partial y} - \tilde{\Gamma}_{ay}^{(PSch)} + \frac{j_y^{(vis\perp)}}{e}, \quad z_{n_a} = 1 \ \& \ z_a = 1, \\ \left(V_{ay}^{(E)} + V_{ay}^{(AN)} \right) n_a - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \frac{1}{h_y} \frac{\partial p_a}{\partial y} - \tilde{\Gamma}_{ay}^{(PSch)}, \end{cases}$$

$$\Gamma_{ax}^{(PSch)} = D_{axy} \frac{1}{h_y} \frac{\partial n_a T_i}{\partial y}$$

$$\Gamma_{ay}^{(PSch)} = -D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i}{\partial x}$$

$$\tilde{\Gamma}_{ax}^{(PSch)} = \begin{cases} \Gamma_{ax}^{(PSch)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{\Gamma_{ax}^{(PSch)}}{1 + \frac{|\Gamma_{ax}^{(PSch)}|}{\frac{\sqrt{g}}{h_x} n_a V_{T_a}}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases}$$

$$\tilde{\Gamma}_{ay}^{(PSch)} = \begin{cases} \Gamma_{ay}^{(PSch)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{\Gamma_{ay}^{(PSch)}}{1 + \frac{|\Gamma_{ay}^{(PSch)}|}{\frac{\sqrt{g}}{h_y} n_a V_{T_a}}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases}$$

$$V_{T_a} = |b_x| \sqrt{\frac{8T_i}{\pi m_a}}$$

$$\Gamma_{ax} = \begin{cases} \left(b_x V_{\parallel a} + \tilde{U}_{xa}^{(dia)} + V_{ax}^{(AN)} + V_a^{corr-dpc} + 0 \cdot \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis||)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{en_a} \right) n_a - \\ - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x}, \quad z_{n_a} = 1 \ \& \ z_a = 1 \\ \left(b_x V_{\parallel a} + \tilde{U}_{xa}^{(dia)} + V_{ax}^{(AN)} + V_a^{corr-dpc} \right) n_a - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x}, \end{cases}$$

$$\Gamma_{ay} = \begin{cases} \left(\tilde{U}_{ya}^{(dia)} + V_{ay}^{(AN)} + \frac{j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + \tilde{j}_y^{(visq)}}{en_a} \right) n_a - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \frac{1}{h_y} \frac{\partial p_a}{\partial y} + \frac{j_y^{(vis\perp)}}{e}, \\ \left(\tilde{U}_{ya}^{(dia)} + V_{ay}^{(AN)} \right) n_a - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \frac{1}{h_y} \frac{\partial p_a}{\partial y} \end{cases} \quad z_{n_a}=1 \text{ \& } z_a=1$$

$$\Gamma_{ax}^{Cor} = \begin{cases} \left(b_x V_{||a} + \tilde{U}_{xa}^{(2dia)} + V_{ax}^{(AN)} + V_a^{corr-dpc} + 0 \cdot \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis||)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{en_a} \right) n_a - \\ - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x}, \quad z_{n_a}=1 \text{ \& } z_a=1 \\ \left(b_x V_{||a} + \tilde{U}_{xa}^{(2dia)} + V_{ax}^{(AN)} + V_a^{corr-dpc} \right) n_a - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} - \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x} \end{cases}$$

$$\Gamma_{ay}^{Cor} = \begin{cases} \left(\tilde{U}_{ya}^{(2dia)} + V_{ay}^{(AN)} + \frac{j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + \tilde{j}_y^{(visq)}}{en_a} \right) n_a - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \frac{1}{h_y} \frac{\partial p_a}{\partial y} + \frac{j_y^{(vis\perp)}}{e}, \quad z_{n_a}=1 \text{ \& } z_a=1 \\ \left(\tilde{U}_{ya}^{(2dia)} + V_{ay}^{(AN)} \right) n_a - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} - \tilde{D}_{ay}^p \frac{1}{h_y} \frac{\partial p_a}{\partial y} \end{cases}$$

$$V_{ax}^{(E)} = \begin{cases} -\frac{B_z}{B^2} \frac{1}{h_y} \frac{\partial \phi}{\partial y}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}, \quad V_{ay}^{(E)} = \begin{cases} \frac{B_z}{B^2} \frac{1}{h_x} \frac{\partial \phi}{\partial x}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases},$$

$$\tilde{V}_{xa}^{(dia)} = \begin{cases} \frac{T_i B_z}{Z_a e} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right), & z_a \neq 0, \\ 0, & z_a = 0 \end{cases}, \quad \tilde{V}_{ya}^{(dia)} = \begin{cases} -\frac{T_i B_z}{Z_a e} \frac{\partial}{\partial x} \left(\frac{1}{B^2} \right), & z_a \neq 0, \\ 0, & z_a = 0 \end{cases},$$

$$\tilde{\tilde{V}}_{xa}^{(dia)} = \begin{cases} \tilde{V}_{xa}^{(dia)}, & (z_{n_a}=1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{\tilde{V}_{xa}^{(dia)}}{1 + \frac{|\tilde{V}_{xa}^{(dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases} \quad \tilde{\tilde{V}}_{ya}^{(dia)} = \begin{cases} \tilde{V}_{ya}^{(dia)}, & (z_{n_a}=1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{\tilde{V}_{ya}^{(dia)}}{1 + \frac{|\tilde{V}_{ya}^{(dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases}$$

$$\Gamma_{ax}^{(Eir)} = \begin{cases} \left(b_x V_{||a} + V_{ax}^{(E)} + \tilde{W}_{xa}^{(dia)} + V_{ax}^{(AN)} + V_a^{corr-dpc} + 0 \cdot \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis||)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{en_a} \right) n_a - \\ - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x}, \quad z_{n_a}=1 \text{ \& } z_a=1 \\ \left(b_x V_{||a} + V_{ax}^{(E)} + \tilde{W}_{xa}^{(dia)} + V_{ax}^{(AN)} + V_a^{corr-dpc} \right) n_a - D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} \end{cases}$$

$$\Gamma_{ay}^{(Eir)} = \begin{cases} \left(V_{ay}^{(E)} + \tilde{W}_{ya}^{(dia)} + V_{ay}^{(AN)} + \frac{j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + \tilde{j}_y^{(visq)}}{en_a} \right) n_a - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} + \frac{j_y^{(vis\perp)}}{e}, & z_{n_a} = 1 \text{ \& } z_a = 1 \\ \left(V_{ay}^{(E)} + \tilde{W}_{ya}^{(dia)} + V_{ay}^{(AN)} \right) n_a - D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} & \end{cases}$$

where

$$\begin{aligned} W_{xa}^{(dia)} &= \begin{cases} -\frac{B_z}{Z_a e B^2 n_a} \frac{\partial n_a T_i}{h_y \partial y}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases} & W_{ya}^{(dia)} &= \begin{cases} \frac{B_z}{Z_a e B^2 n_a} \frac{\partial n_a T_i}{h_x \partial x}, & z_a \neq 0, \\ 0, & z_a = 0 \end{cases} \\ \tilde{W}_{xa}^{(dia)} &= \begin{cases} W_{xa}^{(dia)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{W_{xa}^{(dia)}}{1 + \frac{|W_{xa}^{(dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases} & \tilde{W}_{ya}^{(dia)} &= \begin{cases} W_{ya}^{(dia)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{W_{ya}^{(dia)}}{1 + \frac{|W_{ya}^{(dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases} \\ U_{xa}^{(dia)} &= \tilde{V}_{xa}^{(dia)} + V_{ax}^{(E)} & U_{ya}^{(dia)} &= \tilde{V}_{ya}^{(dia)} + V_{ay}^{(E)} \\ U_{xa}^{(2dia)} &= 2\tilde{V}_{xa}^{(dia)} + V_{ax}^{(E)} & U_{ya}^{(2dia)} &= 2\tilde{V}_{ya}^{(dia)} + V_{ay}^{(E)} \\ \tilde{U}_{xa}^{(dia)} &= \begin{cases} U_{xa}^{(dia)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{U_{xa}^{(dia)}}{1 + \frac{|U_{xa}^{(dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases} & \tilde{U}_{ya}^{(dia)} &= \begin{cases} U_{ya}^{(dia)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{U_{ya}^{(dia)}}{1 + \frac{|U_{ya}^{(dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases} \\ \tilde{U}_{xa}^{(2dia)} &= \begin{cases} U_{xa}^{(2dia)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{U_{xa}^{(2dia)}}{1 + \frac{|U_{xa}^{(2dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases} & \tilde{U}_{ya}^{(2dia)} &= \begin{cases} U_{ya}^{(2dia)}, & (z_{n_a} = 1 \wedge z_a \neq 1) \vee z_a = 0 \\ \frac{U_{ya}^{(2dia)}}{1 + \frac{|U_{ya}^{(2dia)}|}{V_{T_a}}}, & (z_{n_a} \neq 1 \vee z_a \neq 1) \wedge z_a \neq 0 \end{cases} \end{aligned}$$

$$E_{k,a} = \frac{m_a}{2} u_{\parallel a}^2$$

Source terms in density equations

$$S_a^n = S_{I_a}^n + S_{I_{a+1}}^n + S_{R_a}^n + S_{R_{a-1}}^n + S_{CX_a}^n + S_{CX_{a-1}}^n,$$

Z_a - the atomic charge of species (a)

z_{n_a} - the nuclear charge of species (a)

$$S_{I_a}^n = \begin{cases} -r_{I_a} n_e n_a, & a=0,...,n_s-2, \quad z_a < z_{a+1} \text{ and } z_{n_a} = z_{n_{a+1}}, \\ 0, & \end{cases}$$

$$S_{I_{a+1}}^n = \begin{cases} r_{I_a} n_e n_a, & a=0,...,n_s-2, \quad z_a < z_{a+1} \text{ and } z_{n_a} = z_{n_{a+1}}, \\ 0 & \end{cases}$$

$$r_{I_a} = \exp\left(r_{0,I_a} + r_{1,I_a} \ln\left(\frac{T_e}{e}\right)\right)$$

$$S_{R_a}^n = \begin{cases} -r_{R_a} n_e n_a, & a=1,...,n_s-1, \quad z_{a-1} < z_a \text{ and } z_{n_{a-1}} = z_{n_a}, \\ 0 & \end{cases}$$

$$S_{R_{a-1}}^n = \begin{cases} r_{R_a} n_e n_a, & a=1,...,n_s-1, \quad z_{a-1} < z_a \text{ and } z_{n_{a-1}} = z_{n_a}, \\ 0, & \end{cases}$$

$$r_{R_a} = \exp\left(r_{0,R_a} + r_{1,R_a} \ln\left(\frac{T_e}{e}\right)\right)$$

$$S_{CX_a}^n = \begin{cases} -r_{CX_a} n_{a_0} n_a, & a=1,...,n_s-1, \quad z_{a-1} < z_a \text{ and } z_{n_{a-1}} = z_{n_a} \quad ??? \\ 0 & \end{cases}$$

$$S_{CX_{a-1}}^n = \begin{cases} r_{CX_a} n_{a_0} n_a, & a=1,...,n_s-1, \quad z_{a-1} < z_a \text{ and } z_{n_{a-1}} = z_{n_a} \quad ??? \\ 0, & \end{cases}$$

$$r_{CX_a} = \exp\left(r_{0,CX_a} + r_{1,CX_a} \ln\left(\frac{m_p T_i}{m_{a_0} e}\right)\right)$$

$$n_e = \sum_{a=0}^{n_s-1} z_a n_a,$$

$$n_i = \sum_{a=0}^{n_s-1} n_a$$

$$p_a = n_a (z_a T_e + T_i),$$

$$p_e = n_e T_e$$

$$V_a^{corr-dpc} = \frac{c^{corr-dpc}}{16} |b_x| \sqrt{\frac{p_a}{m_a n_a}} \frac{\partial^2}{\partial x^2} \left(\frac{1}{p_a} \frac{\partial p_a}{\partial x} \right)$$

$$\begin{cases} D_{axy} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{z_a e}, & z_a \neq 0 \\ D_{axy} = 0, & z_a = 0 \end{cases}, \text{ where } \langle B^2 \rangle = \frac{\iint_{core} B^2 \sqrt{g} dx dy}{\iint_{core} \sqrt{g} dx dy}$$

$$\begin{cases} D_{ayx} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{z_a e}, & z_a \neq 0 \\ D_{ayx} = 0, & z_a = 0 \end{cases}$$

$$D_a^n = D_{ANa}^n$$

$$D_{ANa}^n = \left(c_{D_a^N,0} + c_{D_a^N,1} \frac{10^{20}}{n_e} + c_{D_a^N,2} D_a^f + c_{D_a^N,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \right) f_D(x, y),$$

$$f_D(x, y) = \begin{cases} 1, & z_a = 0 \\ \alpha_{1,D}, & -\delta_{1,D} \leq y \leq 0, \quad x \in Core, \quad z_a \neq 0 \\ \alpha_{2,D}, & 0 \leq y \leq \delta_{2,D}, \quad x \in Core, \\ \alpha_{3,D}, & x, y \in Pr \cup SOL \end{cases}$$

$$a = 0, \quad c_{D_a^N,0} = 0, \quad c_{D_a^N,1} = 0, \quad c_{D_a^N,2} = 0, \quad c_{D_a^N,3} = 0$$

$$a = 1, \quad c_{D_a^N,0} = 0, \quad c_{D_a^N,1} = 0, \quad c_{D_a^N,2} = 1, \quad c_{D_a^N,3} = 0$$

$$D_a^f = c_{D_a^f,0} + c_{D_a^f,1} \frac{10^{20}}{n_i} + \frac{c_{D_a^f,2} \sqrt{\frac{T_i}{m_a}}}{c_{D_a^f,3} n_i + c_{D_a^f,4} n_e},$$

a	$c_{D_a^f,0}$	$c_{D_a^f,1}$	$c_{D_a^f,2}$	$c_{D_a^f,3}$	$c_{D_a^f,4}$	$c_{D_a^f,5}$	$c_{D_a^f,6}$	$c_{D_a^f,7}$
0	10^2	0	1	$5 \cdot 10^{-20}$	$1,5 \cdot 10^{-18}$	0	0	0
1	0,3	0	0	–	–	0	0	0

$$\tilde{D}_{ax}^p = \begin{cases} \frac{D_a^p}{\left(1 + \frac{\left(\frac{\sqrt{g}}{h_x} D_a^p \left| \frac{1}{h_x} \frac{\partial p_a}{\partial x} \right| \right)^{\gamma_1}}{\alpha \frac{\sqrt{g}}{h_x} n_a \frac{V_{T_i}}{4}} \right)^{\frac{1}{\gamma_1}}}, & V_{T_i} = \sqrt{\frac{8T_i}{\pi m_a}}, \quad z_a = 0 \\ D_a^p, & z_a \neq 0 \end{cases}$$

$$\tilde{D}_{ay}^p = \begin{cases} \frac{D_a^p}{\left(1 + \frac{\left(\frac{\sqrt{g}}{h_y} D_a^p \left| \frac{1}{h_y} \frac{\partial p_a}{\partial y} \right| \right)^{\gamma_1}}{\alpha \frac{\sqrt{g}}{h_y} n_a \frac{V_{T_i}}{4}} \right)^{\frac{1}{\gamma_1}}}, & V_{T_i} = \sqrt{\frac{8T_i}{\pi m_a}}, \quad z_a = 0 \\ D_a^p, & z_a \neq 0 \end{cases}$$

$$D_a^p = D_{ANa}^p$$

$$D_{ANa}^p = \frac{1}{z_a T_e + T_i} \left(c_{D_a^p,0} + c_{D_a^p,1} \frac{10^{20}}{n_e} + c_{D_a^p,2} D_a^f + c_{D_a^p,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \right),$$

$$a=0, c_{D_a^p,0}=0, c_{D_a^p,1}=0, c_{D_a^p,2}=1, c_{D_a^p,3}=0$$

$$a=1, c_{D_a^p,0}=0, c_{D_a^p,1}=0, c_{D_a^p,2}=0, c_{D_a^p,3}=0$$

$$V_{ax}^{(AN)}=0$$

$$V_{ay}^{(AN)}=c_{V_a^{(AN)},0}+c_{V_a^{(AN)},1}\frac{10^{20}}{n_e}+c_{V_a^{(AN)},2}D_a^f+c_{V_a^{(AN)},3}\frac{6,2\cdot 10^{-2}T_e}{e|B|}$$

$$a=0, c_{V_a^{(AN)},0}=0, c_{V_a^{(AN)},1}=0, c_{V_a^{(AN)},2}=0, c_{V_a^{(AN)},3}=0$$

$$a=1, c_{V_a^{(AN)},0}=0, c_{V_a^{(AN)},1}=0, c_{V_a^{(AN)},2}=0, c_{V_a^{(AN)},3}=0$$

New form

$$\begin{aligned}
& m_a \frac{\partial n_a V_{\parallel a}}{\partial t} + \frac{1}{h_z \sqrt{g}} \frac{\partial}{\partial x} \left(\frac{h_z \sqrt{g}}{h_x} \Gamma_{ax}^m \right) + \frac{1}{h_z \sqrt{g}} \frac{\partial}{\partial y} \left(\frac{h_z \sqrt{g}}{h_y} \Gamma_{ay}^m \right) + \frac{b_x}{h_x} \frac{\partial n_a T_i}{\partial x} + Z_a e n_a \frac{b_x}{h_x} \frac{\partial \varphi}{\partial x} = \\
& = S_{a\parallel}^m + S_{CF_a}^m + S_{fr_a}^m + S_{Term_a}^m + S_{I_a}^m + S_{R_a}^m + S_{CX_a}^m + S_{AN_a}^m
\end{aligned}$$

Old form

$$\begin{aligned}
& m_a \frac{\partial n_a V_{\parallel a}}{\partial t} + \frac{1}{h_z \sqrt{g}} \frac{\partial}{\partial x} \left(\frac{h_z \sqrt{g}}{h_x} \Gamma_{ax}^m \right) + \frac{1}{h_z \sqrt{g}} \frac{\partial}{\partial y} \left(\frac{h_z \sqrt{g}}{h_y} \Gamma_{ay}^m \right) + \frac{b_x}{h_x} \frac{\partial n_a T_i}{\partial x} + \frac{b_x}{h_x} \frac{Z_a n_a}{n_e} \frac{\partial T_e}{\partial x} = \\
& = S_{a\parallel}^m + S_{CF,a}^m + S_{fr_a}^m + S_{Term_a}^m + S_{I_a}^m + S_{R_a}^m + S_{CX_a}^m \quad \text{old form}
\end{aligned}$$

$$S_{a\parallel}^m = -(\nabla \pi_a^{\parallel})_{\parallel}$$

$$(\nabla \pi_a^{\parallel})_{\parallel} = \begin{cases} 0, \\ -\frac{\partial}{h_z \sqrt{g} \partial x} \left(\frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \tilde{\eta}_{CLax} \frac{\partial \ln \left(h_z B^{\frac{1}{2}} \right)}{B^{\frac{1}{2}} h_x \partial x} \right) B^{\frac{1}{2}} V_{\parallel a} - B^{\frac{3}{2}} b_x \frac{\partial}{h_x \partial x} \left(\alpha^{NEO} \frac{b_x \tilde{\tau}_a}{B^2} \frac{\partial \left(B^{\frac{1}{2}} \tilde{q}_a^{(0)} \right)}{h_x \partial x} \right), \end{cases} \quad z_{n_a} = 1 \text{ \& } z_a = 1$$

$$S_{CF,a}^m = m_a b_x n_a V_{\parallel a}^2 \frac{1}{h_z h_x} \frac{\partial h_z}{\partial x}$$

$$S_{AN_a}^m = \begin{cases} 0, & z_a \neq 1 \text{ or } z_{n,a} \neq 1 \\ -j_y^{(AN)} \frac{B B_x}{B_z}, & z_a = 1 \text{ and } z_{n,a} = 1 \end{cases} \quad \text{-correcting source due to artificial anomalous electric conductivity}$$

$$S_a^m = S_{fr,a}^m + S_{Term}^m + S_{I_a}^m + S_{R_a}^m + S_{CX_a}^m,$$

Parallel momentum source due to ionization

$$S_{I_a}^m = S_{I_a}^q + S_{I_{a+1}}^q$$

$$S_{I_a}^q = \begin{cases} -r_{I_a} n_e n_a \left(\frac{m_a + m_{a+1}}{2} \right) V_{\parallel a}, & a = 0, \dots, n_s - 2, \quad z_a < z_{a+1} \text{ and } z_{n_a} = z_{n_{a+1}}, \\ 0, & \end{cases}$$

$$S_{I_{a+1}}^q = \begin{cases} r_{I_a} n_e n_a \left(\frac{m_a + m_{a+1}}{2} \right) V_{\parallel a}, & a = 0, \dots, n_s - 2, \quad z_a < z_{a+1} \text{ and } z_{n_a} = z_{n_{a+1}}, \\ 0, & \end{cases}$$

Parallel momentum source due to recombination

$$S_{R_a}^m = S_{R_a}^q + S_{R_{a-1}}^q$$

$$S_{R_a}^q = \begin{cases} -r_{R_a} n_e n_a \left(\frac{m_a + m_{a-1}}{2} \right) V_{\parallel a}, & a=1, \dots, n_s-1, \quad z_{a-1} < z_a \quad \text{and} \quad z_{n_{a-1}} = z_{n_a} \\ 0 & \end{cases}$$

$$S_{R_{a-1}}^q = \begin{cases} r_{R_a} n_e n_a \left(\frac{m_a + m_{a-1}}{2} \right) V_{\parallel a}, & a=1, \dots, n_s-1, \quad z_{a-1} < z_a \quad \text{and} \quad z_{n_{a-1}} = z_{n_a}, \\ 0, & \end{cases}$$

$$S_{CX_a}^m =$$

New friction

$$S_{fr_a}^m = S_{fr,ea}^m + S_{fr,ia}^m$$

$$S_{fr,ea}^m = \frac{b_x^2 e^2 z_a^2}{\sigma_{\parallel} z_{eff}} n_e n_a (V_{e\parallel} - V_{a\parallel}) \quad z_a \neq 0$$

$$S_{fr,ia}^m = \sum_{\substack{b=0 \\ b \neq a}}^{n_s-1} R_{ab}^{(V_{\parallel})} = \sum_{\substack{b=0 \\ b \neq a}}^{n_s-1} \frac{1}{\zeta_p} m_p \sqrt{\frac{\frac{m_a}{m_p} \frac{m_b}{m_p}}{\frac{m_a}{m_p} + \frac{m_b}{m_p}}} z_a^2 z_b^2 n_a n_b K_{ab}^{(V_{\parallel})} (V_{b\parallel} - V_{a\parallel})$$

$$\zeta_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_p} T_i^{\frac{3}{2}}}{\Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2 \quad \tau_a = \frac{\zeta_p}{z_a^2 z_{eff} n_e} \sqrt{\frac{m_a}{m_p}}$$

$$K_{rs}^{(V_{\parallel})} = \begin{cases} c_{imp}^{(1)}, & r = r_{main}, \quad s \neq r, \quad z_r \neq 0, \quad z_s \neq 0, \\ c_{imp}^{(1)}, & s = s_{main}, \quad r \neq s, \quad z_r \neq 0, \quad z_s \neq 0, \\ 1, & r \neq r_{main}, \quad s \neq s_{main}, \quad z_r \neq 0, \quad z_s \neq 0, \\ 0, & r = s, \quad \text{or} \quad z_r = 0, \quad z_s = 0, \end{cases}$$

$$c_{imp}^{(1)} = \frac{(1+0.24z_{eff_imp})(1+0.93z_{eff_imp})}{(1+2.56z_{eff_imp})(1+0.29z_{eff_imp})} \quad c_{imp}^{(2)} = 1.56 \frac{(1+1.4z_{eff_imp})(1+0.52z_{eff_imp})}{(1+2.56z_{eff_imp})(1+0.29z_{eff_imp})}$$

$$z_{eff_imp} = \frac{\sum_{\substack{a=0 \\ a \neq a_{main}}}^{n_s-1} n_a z_a^2}{n_{a_{main}} z_{a_{main}}^2}$$

Old friction

$$S_{fr,a}^m = \sum_{\substack{b=0 \\ b \neq a, z_a \neq 0, z_b \neq 0}}^{n_s-1} \frac{1}{\tau_p} m_p \sqrt{\frac{\frac{m_a}{m_p} \frac{m_b}{m_p}}{\frac{m_a}{m_p} + \frac{m_b}{m_p}}} z_a^2 z_b^2 n_b n_a (V_{b\parallel} - V_{a\parallel}), \text{ - friction force (b2sifr)}$$

$$\tau_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_p} T_i^{\frac{3}{2}}}{\Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2, \text{ where the Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23,4 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 3,45 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} \leq 50 \\ \max \left\{ 5; 25,3 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 2,30 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} > 50 \\ \max \left\{ 5; 17,3 - 0,5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) + 1,5 \cdot \log_e \left(\frac{T_e}{e \cdot 10^3} \right) \right\} \end{cases} \quad \begin{array}{l} \text{Braginsky formula} \\ \\ \text{Wesson formula for ion-ion collisions} \end{array},$$

New thermal force

$$S_{Term,a}^m = S_{Term,ea}^m + S_{Term,ia}^m$$

$$S_{Term,ea}^m = -\alpha_{ex} c_e^{(1)} \frac{n_a z_a^2 m_e}{b_x e \zeta_e} \frac{1}{c_{\sigma_{CL}}^{(Luciani)}} \frac{1}{h_x} \frac{\partial T_e}{\partial x}, \quad z_a \neq 0$$

$$S_{Term,ia}^m = \sum_{\substack{b=0 \\ b \neq a}}^{n_s-1} c_{\alpha_{ab}}^{(F\lim)} z_a^2 z_b^2 \frac{m_p}{\zeta_p e} \frac{n_a n_b}{n_a + n_b} \alpha_{ab} \frac{1}{h_x} \frac{\partial T_i}{\partial x}, \quad z_a \neq 0$$

$$\alpha_{ab} = \frac{\zeta_p e (n_a + n_b)}{\sqrt{m_p}} \sqrt{\frac{\frac{m_a}{m_p} \frac{m_b}{m_p}}{\frac{m_a}{m_p} + \frac{m_b}{m_p}}} \left(\frac{K_{ab}^{(Term)}}{\kappa_b} - \frac{K_{ba}^{(Term)}}{\kappa_a} \right) b_x, \quad z_a \neq 0$$

$$\zeta_e = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_e T_e^2}}{\Lambda} \left(\frac{4\pi \varepsilon_0}{e^2} \right)^2 \quad \tau_e = \frac{\zeta_e}{z_{eff} n_e}$$

$$c_e^{(1)} = \frac{(1 + 0.24 z_{eff})(1 + 0.93 z_{eff})}{(1 + 2.56 z_{eff})(1 + 0.29 z_{eff})}, \quad c_e^{(2)} = 1.56 \frac{(1 + 1.4 z_{eff})(1 + 0.52 z_{eff})}{(1 + 2.56 z_{eff})(1 + 0.29 z_{eff})} \quad \tilde{c}_e^{(2)} = \frac{c_e^{(2)}}{\left(z_{eff} + \frac{\sqrt{2}}{2} \right)}$$

$$K_{rs}^{(Term)} = \begin{cases} c_{imp}^{(2)}, & s \neq r, \quad s = s_{main}, \quad z_r \neq 0, \quad z_s \neq 0, \\ 0, & r \neq r_{main}, \quad s \neq s_{main}, \quad z_r \neq 0, \quad z_s \neq 0, \\ 0, & r = r_{main}, \quad s - \text{arbitrary} \end{cases}$$

$$\kappa_a = \sum_{r=0}^{n_s-1} z_a^2 z_r^2 n_r \sqrt{m_p} \sqrt{\frac{\frac{m_a}{m_p} \frac{m_r}{m_p}}{\frac{m_a}{m_p} + \frac{m_r}{m_p}}}$$

$$c_{\alpha_{ab}}^{(F\lim)} = \begin{cases} 1, & \text{inside the separatrix} \\ \frac{1}{1 + \frac{\alpha_{ab} \frac{1}{h_x} \frac{\partial T_i}{\partial x}}{c_{flimab} j_{ab\max}}}, & \text{outside the separatrix} \end{cases}$$

$$j_{ab\max} = e K_{ab}^{(V_1)} \sqrt{\frac{T_i}{m_p}} (n_a + n_b)$$

New coefficient c_{flimab} is stored in cflim(4)

Old thermal force

$$S_{Term,a}^m = n_a \left(\frac{z_a}{n_e} - \frac{z_a^2}{n_{e^2}} \right) n_e b_x e \left[\hat{E}_x - \frac{1}{e} (c_{th_e,a} \Delta T_e + c_{th_i,a} \Delta T_i) \right], \text{ thermal force (b2sifr)}$$

$$n_{e^2} = \sum_{a=0}^{n_s-1} z_a^2 n_a, \quad l_{t_e} = 0,3; \quad l_{t_i} = 0,3; \quad z_{\lambda_e} = \frac{1.5 \cdot 10^{16}}{n_e} \left(\frac{T_e}{e} \right)^2, \quad z_{\lambda_i} = \frac{1.5 \cdot 10^{16}}{n_{e^2}} \left(\frac{T_i}{e} \right)^2$$

$$\Delta T_e = \begin{cases} \frac{\partial T_e}{h_x \partial x}, & l_{t_e} \leq 0 \\ \left| \min \left\{ \left| \frac{\partial T_e}{h_x \partial x} \right|, \frac{l_{t_e} T_e}{z_{\lambda_e} |b_x|} \right\} \right|, & \frac{\partial T_e}{h_x \partial x} \geq 0, \quad l_{t_e} > 0 \\ - \left| \min \left\{ \left| \frac{\partial T_e}{h_x \partial x} \right|, \frac{l_{t_e} T_e}{z_{\lambda_e} |b_x|} \right\} \right|, & \frac{\partial T_e}{h_x \partial x} < 0, \quad l_{t_e} > 0 \end{cases}$$

$$\Delta T_i = \begin{cases} \frac{\partial T_i}{h_x \partial x}, & l_{t_i} \leq 0 \\ \left| \min \left\{ \left| \frac{\partial T_i}{h_x \partial x} \right|, \frac{l_{t_i} T_i}{z_{\lambda_i} |b_x|} \right\} \right|, & \frac{\partial T_i}{h_x \partial x} \geq 0, \quad l_{t_i} > 0 \\ - \left| \min \left\{ \left| \frac{\partial T_i}{h_x \partial x} \right|, \frac{l_{t_i} T_i}{z_{\lambda_i} |b_x|} \right\} \right|, & \frac{\partial T_i}{h_x \partial x} < 0, \quad l_{t_i} > 0 \end{cases}$$

$c_{th_e,a} = 0, a = 0, n_s - 1$ and $c_{th_i,a} = 2,65, a = 0, n_s - 1$ if 21 moment approximation is not used.

$$\Gamma_{ax}^m = \begin{cases} m_a V_{\parallel a} \Gamma_{ax}^{Cor} + \frac{4}{3} \tilde{\eta}_{CLax} \frac{\partial \ln h_z}{h_x \partial x} V_{\parallel a} - \eta_{ax} \frac{\partial \mathcal{V}_{\parallel a}}{h_x \partial x}, & z_{n_a} = 1 \text{ \& } z_a = 1 \\ m_a V_{\parallel a} \Gamma_{ax}^{Cor} - \eta_{ax} \frac{\partial \mathcal{V}_{\parallel a}}{h_x \partial x}, & \end{cases}$$

$$\Gamma_{ay}^m = m_a V_{\parallel a} \Gamma_{ay}^{Cor} - \eta_{ay} \frac{\partial \mathcal{V}_{\parallel a}}{h_y \partial y},$$

$$\eta_{ax} = \eta_{ANa} + \frac{4}{3} \tilde{\eta}_{CLax}$$

$$\eta_{ANa} = \begin{cases} \left(c_{\eta_a,0} + c_{\eta_a,1} \frac{10^{20}}{n_e} + c_{\eta_a,2} D_a^f + c_{\eta_a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \right) m_a n_a f_{\eta_a}(x,y), & z_a \neq 0 \\ \left(c_{\eta_a,0} + c_{\eta_a,1} \frac{10^{20}}{n_e} + c_{\eta_a,2} D_a^f + c_{\eta_a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \right) m_a n_a, & z_a = 0 \end{cases}$$

$$f_{\eta_a}(x, y) = \begin{cases} 1, & z_a = 0 \\ \alpha_{1, \eta_a}, & -\delta_{1, \eta_a} \leq y \leq 0, \quad x \in \text{Core}, \quad z_a \neq 0 \\ \alpha_{2, \eta_a}, & 0 \leq y \leq \delta_{2, \eta_a}, \quad x \in \text{Core}, \\ \alpha_{3, \eta_a}, & x, y \in \text{Pr} \cup \text{SOL} \end{cases}$$

$$a = 0, \quad c_{\eta_a, 0} = 0, \quad c_{\eta_a, 1} = 0, \quad c_{\eta_a, 2} = 1, \quad c_{\eta_a, 3} = 0$$

$$a = 1, \quad c_{\eta_a, 0} = 0, \quad c_{\eta_a, 1} = 0, \quad c_{\eta_a, 2} = 1, \quad c_{\eta_a, 3} = 0$$

$$\tilde{\eta}_{CLax} = c_{\eta_{ax}^{(CL)}} \tilde{\eta}_{CLax}$$

$$c_{\eta_{ax}^{(CL)}} = \begin{cases} 1, & z_a \neq 0 \text{ and only for inner flux surfaces} \\ \frac{1}{1 + \frac{\frac{\sqrt{g}}{h_x} \tilde{\eta}_{CLax} \left| \frac{\partial V_{||a}}{h_x \partial x} \right|}{V_{m_a}^{(\max)}}}}, & \text{for all others cases} \end{cases}$$

$$V_{m_a}^{(\max)} = c_{f \lim_2} \frac{\sqrt{g}}{h_x} |b_x| n_a T_i$$

$$\tilde{\eta}_{CLax} = c_{\eta_{ax}^{(CL)}}^{(Luciani)} \eta_{CLax}$$

'b2trcl_lluciani' '1'

$$c_{\eta_{ax}^{(CL)}}^{(Luciani)} = \begin{cases} \frac{1}{7,5 \cdot 10^{16} \left(\frac{T_i}{e} \right)^2}, & z_a \neq 0 \text{ and only for inner flux surfaces,} \\ 1 + \frac{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}, & \\ 1, & \text{for all others cases} \end{cases}$$

For plateau and banana correction 'b2trcl_lluciani' '3'

$$c_{\eta_{ax}^{(CL)}}^{(Luciani)} = \begin{cases} \frac{1}{15,12 \cdot 10^{16} \left(\frac{T_i}{e} \right)^2} \cdot \frac{1}{15,12 \cdot 10^{16} \cdot \varepsilon^{3/2} \left(\frac{T_i}{e} \right)^2}, & z_a \neq 0 \text{ and only for inner flux surfaces,} \\ 1 + \frac{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}, & \\ 1, & \text{for all others cases} \end{cases}$$

$$c(y) = \oint h_x |b_x|^{-1} dx, \quad \varepsilon = \frac{\left(\oint h_x dx \right)^2}{\oint \sqrt{g} / h_y dx}$$

$$\eta_{CLax} = \begin{cases} 0, & z_a = 0 \\ \eta \sqrt{2T_i n_a \tau_a b_x^2}, & z_a \neq 0 \end{cases}, \eta = 0.96,$$

Old expression was $\eta_{CLax} = \begin{cases} 0, & z_a = 0 \\ \eta T_i n_a \tau_a b_x^2, & z_a \neq 0 \end{cases}, \eta = 1.357,$

$$\tau_a = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_a T_i^{\frac{3}{2}}}}{z_a^2 z_{eff} n_e \Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2, \text{ where the Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23.4 - 1.15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 3.45 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} \leq 50 \\ \max \left\{ 5; 25.3 - 1.15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 2.30 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} > 50 \\ \max \left\{ 5; 17.3 - 0.5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) + 1.5 \cdot \log_e \left(\frac{T_e}{e \cdot 10^3} \right) \right\} \end{cases},$$

Braginsky formula

Wesson formula for ion-ion collisions

$$\eta_{ay} = \eta_{ANa} + \eta_{CLay}, \quad \eta_{ay}^{(CL)} = 0$$

$$\tilde{q}_a^{(0)} = q^{(1)} \arctg \left(\frac{q_a^{(0)}}{q^{(1)}} \right), \quad q^{(1)} = c_{q1} n_i T_i \sqrt{\frac{T_i + T_e}{m_i}}$$

$$c_{q1} = 0.5 \quad \text{or} \quad \{ 'b2siav_cqip1' \quad '0.1' \} \text{ in b2mn.dat}$$

the three-point smoothing procedure is imposed twice on $\tilde{q}^{(0)}$ along x-direction

in case if $\{ 'b2siav_style_qip' \quad '0' \}$ in b2mn.dat

$$q_a^{(0)} = q_{a||}^{(0)} + \frac{B}{B_x} q_{ax}^{(dia)} = \begin{cases} 0 & \text{outside the separatrix} \\ -\frac{5}{2} \frac{B_z B}{B_x < B^2 >} \frac{n_a T_i}{e} \frac{\partial T_i}{h_y \partial y} & \text{inside the separatrix} \end{cases}$$

in case if $\{ 'b2siav_style_qip' \quad '1' \}$ in b2mn.dat

$$q_a^{(0)} = q_{a||}^{(0)} + \frac{B}{B_x} q_{ax}^{(dia)} = \begin{cases} 0 & \text{outside the separatrix} \\ -\tilde{k}_{ix}^{(CL)} b_x \frac{\partial T_i}{h_x \partial x} - \frac{5}{2} \frac{B_z B}{B_x B^2} \frac{n_a T_i}{e} \frac{\partial T_i}{h_y \partial y} & \text{inside the separatrix} \end{cases}$$

$$\tilde{\tau}_a = \frac{\tau_a}{1 + \frac{15,12 \cdot 10^{16} \left(\frac{T_i}{e} \right)^2}{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}} \cdot \frac{1}{1 + \frac{15,12 \cdot 10^{16} \cdot \epsilon^{3/2} \left(\frac{T_i}{e} \right)^2}{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}}$$

This correction is compatible **only** with plateau and banana correction 'b2trcl_lluciani' '3'

$\alpha^{NEO} = \frac{8}{15} \left(\frac{-0.17 + 1.05\nu_*^{1/2} + 2.7\nu_*^2\epsilon^3}{1 + 0.7\nu_*^{1/2} + \nu_*^2\epsilon^3} - 1 \right)$ gives the correct coefficient before the temperature gradient in the neoclassical electric field in the plateau and banana regime

$$\nu_* = \frac{c(y)}{2\pi\epsilon^{3/2}\tau_a\sqrt{T_i/m_a}}, \quad c(y) = \oint h_x |b_x|^{-1} dx, \quad \epsilon = \frac{(\oint h_x dx)^2}{\oint \sqrt{g}/h_y dx}$$

Old form

$$\begin{aligned} & \frac{3}{2} \frac{\partial_e T_e}{\partial} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} \tilde{q}_{ex} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \right) + \frac{n_e T_e}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} b_x V_{e\parallel} \right) = \\ & = Q_e + c_{E \times B} n_e T_e B_z \frac{1}{h_x h_y} \left(\frac{\partial \phi}{\partial y} \frac{\partial}{\partial x} \left(\frac{1}{B^2} \right) - \frac{\partial \phi}{\partial x} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \right) - \frac{j_y^{(ST)}}{en_e} \frac{\partial_e T_e}{h_y \partial y} + j_x \hat{E}_x + 0 \cdot j_y \hat{E}_y, \end{aligned} \quad (3)$$

New form is used when b2sigp_style is 2

$$\begin{aligned} & \frac{3}{2} \frac{\partial_e T_e}{\partial} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} \tilde{q}_{ex} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \right) + \frac{n_e T_e}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} b_x V_{e\parallel} \right) = \\ & = Q_e + c_{E \times B} n_e T_e B_z \frac{1}{h_x h_y} \left(\frac{\partial \phi}{\partial y} \frac{\partial}{\partial x} \left(\frac{1}{B^2} \right) - \frac{\partial \phi}{\partial x} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \right) - \frac{j_y^{(ST)}}{en_e} \frac{\partial_e T_e}{h_y \partial y} + Q_{F_{ei}}, \end{aligned} \quad (3)$$

$$Q_{F_{ei}} = V_{e\parallel} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} (S_{fr,ea}^m + S_{Term,ea}^m)$$

$$Q_e = -Q_\Delta - Q_R$$

$$Q_\Delta = c_{eqp} \Lambda e^{\frac{3}{2}} \frac{\sum_{a=0}^{n_s-1} \frac{z_a^2 m_p}{m_a} n_a}{n_i} \frac{n_e n_i}{T_e^{\frac{3}{2}}} (T_e - T_i), \quad c_{eqp} = 4.8 \cdot 10^{-15}, \quad \text{where } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23.4 - 1.15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 3.45 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} \leq 50 \\ \max \left\{ 5; 25.3 - 1.15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 2.30 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} > 50 \\ \max \left\{ 5; 15.2 - 0.5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) + \log_e \left(\frac{T_e}{e \cdot 10^3} \right) \right\} \end{cases} \quad \begin{array}{l} \text{Braginsky formula} \\ \\ \text{Wesson formula for electron - ion collisions} \end{array},$$

$$Q_R = \sum_{a=0}^{n_s-1} r_{Q_a} n_e n_a \text{ from b2stel}$$

$$\tilde{q}_{ex} = \tilde{f}_{ex} T_e - \kappa_{ex} \frac{1}{h_x} \frac{\partial T_e}{\partial x} + 0 \cdot \alpha_{ex} T_e \hat{E}_x - q_{ex}^{P.Sch}$$

New

$$\tilde{f}_{ex} = \frac{3}{2} \Gamma_{ex} + V_{ex}^{(Str)} n_e - \tilde{c}_e^{(2)} \frac{j_x^{(ll)}}{e}$$

Old

$$\tilde{f}_{ex} = \frac{3}{2} \Gamma_{ex} + V_{ex}^{(Str)} n_e - 0.71 \frac{j_x^{(ll)}}{e}$$

$$q_{ex}^{P.Sch} = -\frac{5}{2} D_{exy} \frac{1}{h_y} \frac{\partial n_e T_e^2}{\partial y}$$

$$D_{exy} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{e} \quad D_{eyx} = \left(\frac{1}{B^2} - \frac{1}{\langle B^2 \rangle} \right) \frac{B_z}{e} \quad \text{where } \langle B^2 \rangle = \frac{\iint_{core} B^2 \sqrt{g} dx dy}{\iint_{core} \sqrt{g} dx dy}$$

New

$$q_{ex} = \left[\frac{3}{2} \Gamma_{ex} - \tilde{c}_e^{(2)} \frac{j_x^{(ll)}}{e} - \frac{5}{2} n_e T_e \frac{B_z}{e} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) + V_{ex}^{(Str)} n_e \right] T_e - \kappa_{ex} \frac{1}{h_x} \frac{\partial T_e}{\partial x} + 0 \cdot \alpha_{ex} T_e \hat{E}_x$$

Old

$$q_{ex} = \left[\frac{3}{2} \Gamma_{ex} - 0.71 \frac{j_x^{(ll)}}{e} - \frac{5}{2} n_e T_e \frac{B_z}{e} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) + V_{ex}^{(Str)} n_e \right] T_e - \kappa_{ex} \frac{1}{h_x} \frac{\partial T_e}{\partial x} + 0 \cdot \alpha_{ex} T_e \hat{E}_x$$

$$\Gamma_{ex} = \sum_{a=0}^{n_s-1} z_a \Gamma_{ax}^{(he)} - \frac{j_x^{(ll)}}{e},$$

$$\Gamma_{ax}^{(he)} = (b_x V_{a||} + V_{ax}^{(E)} + \frac{5}{3} V_a^{(AN)} + V_a^{corr-dpc}) n_a - \frac{5}{3} \left[D_a^n \frac{1}{h_x} \frac{\partial n_a}{\partial x} + \tilde{D}_{ax}^p \frac{1}{h_x} \frac{\partial p_a}{\partial x} \right]$$

New

$$q_{ex}^{(Eir)} = \left[\frac{5}{2} \Gamma_{ex}^{(Eir)} + V_{ex}^{(Str)} n_e - \tilde{c}_e^{(2)} \frac{j_{||}^{(ll)}}{e} \right] T_e - \kappa_{ex} \frac{1}{h_x} \frac{\partial T_e}{\partial x}$$

Old

$$q_{ex}^{(Eir)} = \left[\frac{5}{2} \Gamma_{ex}^{(Eir)} + V_{ex}^{(Str)} n_e - 0.71 \frac{j_{||}^{(ll)}}{e} \right] T_e - \kappa_{ex} \frac{1}{h_x} \frac{\partial T_e}{\partial x}$$

$$q_{ey}^{(Eir)} = \left[\frac{5}{2} \Gamma_{ey}^{(Eir)} + V_{ey}^{(Str)} n_e \right] T_e - \kappa_{ey} \frac{1}{h_y} \frac{\partial T_e}{\partial y}$$

where

$$\Gamma_{ex}^{(Eir)} = \left(b_x V_{|e} + V_{ex}^{(E)} + V_{ex}^{(dia)} + V_{ex}^{(AN)} \right) n_e - D_a^n \frac{1}{h_x} \frac{\partial n_e}{\partial x}$$

$$\Gamma_{ey}^{(Eir)} = \left(V_{ey}^{(E)} + V_{ey}^{(dia)} + V_{ey}^{(AN)} \right) n_e - D_a^n \frac{1}{h_y} \frac{\partial n_e}{\partial y}$$

$$V_{ex}^{(dia)} = \frac{B_z}{eB^2 n_e} \frac{\partial n_e T_e}{h_y \partial y} \quad V_{ey}^{(dia)} = -\frac{B_z}{eB^2 n_e} \frac{\partial n_e T_e}{h_x \partial x}$$

$$V_{ex}^{(Str)} = 0$$

$$\kappa_{ex} = \kappa_e^{(AN)} + \tilde{\kappa}_{ex}^{(CL)},$$

$$\kappa_e^{(AN)} = \chi_e n_e,$$

$$\chi_e = \left(c_{h_e,0} + c_{h_e,1} \frac{10^{20}}{n_e} + c_{h_e,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \right) f_{h_e}(x, y),$$

$$c_{h_e,0} = 1.2, c_{h_e,1} = 0, c_{h_e,3} = 0,$$

$$f_{h_e}(x, y) = \begin{cases} \alpha_{1,h_e}, & -\delta_{1,h_e} \leq y \leq 0, \quad x \in Core, \\ \alpha_{2,h_e}, & 0 \leq y \leq \delta_{2,h_e}, \quad x \in Core, \\ \alpha_{3,h_e}, & x, y \in Pr \cup SOL \end{cases}$$

$$\tilde{\kappa}_{ex}^{(CL)} = c_{k_{ex}}^{(Flim)} \tilde{\kappa}_{ex}^{(CL)}$$

$$c_{k_{ex}}^{(Flim)} = \begin{cases} 1, & \text{only for inner flux surfaces} \\ \frac{1}{1 + \frac{\frac{\sqrt{g}}{h_x} \tilde{\kappa}_{ex}^{(CL)} \left| \frac{\partial T_e}{h_x \partial x} \right|}{V_{h_e}^{(max)} n_e T_e}}, & \text{for other flux surfaces} \end{cases}$$

$$V_{h_e}^{(max)} = c_{flim_0} \frac{\sqrt{g}}{h_x} |b_x| \sqrt{\frac{T_e}{m_e}},$$

$$\tilde{\kappa}_{ex}^{(CL)} = c_{\kappa_{ex}^{(CL)}}^{(Luciani)} \kappa_{ex}^{(CL)} b_x^2,$$

For plateau and banana correction ‘b2trcl_lluciani’ ‘3’

$$c_{\kappa_{ex}^{(CL)}}^{(Luciani)} = \begin{cases} 0.58 \varepsilon^{3/2} / K_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

$$K_1 = \frac{1}{1 + 2.01 \cdot \nu_{*e}^{1/2} + 1.53 \cdot \nu_{*e}} + \frac{0.52 \cdot \varepsilon^3 \cdot \nu_{*e}}{1 + 0.89 \cdot \varepsilon^{3/2} \cdot \nu_{*e}}$$

$$v_{*e} = \frac{c(y)}{2\pi\epsilon^{3/2}\tau_e\sqrt{T_e/m_e}},$$

$$\tau_e = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_e T_e^2}}{z_{eff} n_e \Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2, \text{ where Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 \\ \max \left\{ 5; 23,4 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 3,45 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} \leq 50 \\ \max \left\{ 5; 25,3 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 2,30 \cdot \log_{10} \left(\frac{T_e}{e} \right) \right\}, & \frac{T_e}{e} > 50 \\ \max \left\{ 5; 14,9 - 0,5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) + \log_e \left(\frac{T_e}{e \cdot 10^3} \right) \right\} \end{cases} \quad \begin{array}{l} \text{Braginsky formula} \\ \\ \text{Wesson formula for electron - electron collisions} \end{array},$$

but it was as below

$$c_{\kappa_{ex}^{(CL)}}^{(Luciani)} = \begin{cases} \frac{1}{7,5 \cdot 10^{16} \left(\frac{T_e}{e} \right)^2}, & \text{only for inner flux surfaces} \\ 1 + \frac{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}, & \text{for other flux surfaces} \\ 1, & \end{cases}$$

$$c(y) = \begin{cases} \int \frac{h_x(x, y)}{|b_x|} dx, & \text{for inner flux surfaces} \\ 0, & \text{for other} \end{cases}$$

$$\kappa_{ex}^{(CL)} = \frac{5}{2} \frac{T_e}{m_e} f_{k_e}(z_{eff}) n_e \tau_e, \text{ old form}$$

$$f_{k_e}(z_{eff}) = \frac{2,16 z_{eff}}{1 + 0,27 z_{eff}}, \text{ old form}$$

$$\kappa_{ex}^{(CL)} = \frac{T_e}{m_e} f_{k_e}(z_{eff}) n_e \tau_e, \text{ new form}$$

$$f_{k_e}(z_{eff}) = \frac{3,9 + 2,3/z_{eff}}{0,31 + 1,2/z_{eff} + 0,41/z_{eff}^2}, \text{ new Zhdanov's expression}$$

$$z_{eff} = \frac{\sum_{a=0}^{n_s-1} z_a^2 n_a}{n_e},$$

$$V_{e||} = \frac{1}{en_e} \left[\sum_{a=0}^{n_s-1} z_a en_a V_{a||} - \frac{j_x^{(||)}}{b_x} \right],$$

New expression of parallel current

$$j_x^{(||)} = \sigma_{||} \left(\frac{1}{en_e h_x} \frac{\partial n_e T_e}{\partial x} - \frac{1}{h_x} \frac{\partial \phi}{\partial x} \right) - \alpha_{ex} \frac{1}{h_x} \frac{\partial T_e}{\partial x} - \frac{eb_x}{z_{eff}} \sum_{a=0}^{n_s-1} n_a V_{a||} (z_a^2 - z_a z_{eff})$$

Old

$$j_x^{(||)} = \sigma_{||} \hat{E}_x - \alpha_{ex} \frac{1}{h_x} \frac{\partial T_e}{\partial x},$$

$$\hat{E}_x = \frac{1}{en_e h_x} \frac{\partial n_e T_e}{\partial x} - \frac{1}{h_x} \frac{\partial \phi}{\partial x},$$

$$\sigma_{||} = \sigma_{AN||} + \tilde{\sigma}_{CL||},$$

$$\sigma_{AN||} = 0$$

$$\tilde{\sigma}_{CL||} = c_{\sigma_{CL||}}^{(Luciani)} \sigma_{CL||}$$

For plateau and banana correction 'b2trcl_lluciani' '3'

$$c_{\sigma_{CL||}}^{(Luciani)} = \begin{cases} 0.58 \varepsilon^{3/2} / K_1, & \text{only for inner flux surfaces} \\ 1, & \text{for other flux surfaces} \end{cases}$$

Old form

$$c_{\sigma_{CL||}}^{(Luciani)} = \frac{1}{1 + \frac{7.5 \cdot 10^{16} \left(\frac{T_e}{e} \right)^2}{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}}$$

New form

$$\sigma_{CL||} = b_x^2 \frac{e^2}{m_e} f_{\sigma_e}(z_{eff}) n_e \tau_e, \quad f_{\sigma_e}(z_{eff}) = \frac{1}{c_e^{(1)} z_{eff}}$$

Old form

$$\sigma_{CL||} = b_x^2 \frac{e^2}{m_e} f_{\sigma_e}(z_{eff}) n_e \tau_e, \quad f_{\sigma_e}(z_{eff}) = \frac{4.95 z_{eff}}{1 + 1.54 z_{eff}}$$

$$\alpha_{ex} = \alpha_{ANx} + \tilde{\alpha}_{CLx},$$

$$\alpha_{ANx} = 0,$$

$$\tilde{\alpha}_{CLx} = c_{k_e}^{(Flim)} \tilde{\alpha}_{CLx}$$

$$\tilde{\alpha}_{CLx} = c_{k^{(CL)}_{ex}}^{(Luciani)} \alpha_{CLx}$$

New form

$$\tilde{\alpha}_{CLx} = c_{\alpha_{ex}}^{(F\lim)} \tilde{\alpha}_{CLx}, \quad \tilde{\alpha}_{CLx} = c_{\alpha_{ex}}^{(Luciani)} \alpha_{CLx}, \quad \alpha_{CLx} = b_x^2 \frac{e}{m_e} f_{\alpha}(z_{eff}) \tau_e n_e z_{eff}, \quad f_{\alpha}(z_{eff}) = -\frac{\tilde{c}_e^{(2)}}{c_e^{(1)}}$$

$$c_{\alpha_{ex}}^{(F\lim)} = \begin{cases} 1, & \text{inside the separatrix} \\ \frac{1}{1 + \frac{\alpha_{CLx} \frac{1}{h_x} \frac{\partial T_e}{\partial x}}{c_{flime} j_{e\max}}}, & \text{outside the separatrix} \end{cases}$$

$$j_{e\max} = |b_x| e n_e \sqrt{\frac{T_e}{m_e}}$$

New coefficient c_{flime} is stored in cflim(3)

Old form

$$\alpha_{CLx} = b_x^2 \sqrt{\frac{5}{2}} \frac{e}{m_e} f_{\alpha}(z_{eff}) n_e \tau_e, \quad f_{\alpha}(z_{eff}) = -\frac{1,25 z_{eff}}{1 + 0,41 z_{eff}},$$

$$\tilde{q}_{ey} = \tilde{f}_{ey} T_e - \kappa_{ey} \frac{1}{h_y} \frac{\partial T_e}{\partial y} + 0 \cdot \alpha_{ey} T_e \hat{E}_y - q_{ey}^{P.Sch}$$

$$\tilde{f}_{ey} = \frac{3}{2} \left(\Gamma_{ey} - \frac{5}{3} \frac{j_y^{(ST)}}{e} \right) + V_{ey}^{(Str)} n_e$$

$$q_{ey}^{P.Sch} = \frac{5}{2} D_{eyx} \frac{1}{h_x} \frac{\partial n_e T_e^2}{\partial x}$$

$$q_{ey} = \left[\frac{3}{2} \Gamma_{ey} - \frac{5}{2} \frac{j_y^{(ST)}}{e} + \frac{5}{2} n_e T_e \frac{B_z}{e} \frac{\partial}{\partial x} \left(\frac{1}{B^2} \right) + V_{ey}^{(Str)} n_e \right] T_e - \kappa_{ey} \frac{1}{h_y} \frac{\partial T_e}{\partial y} + 0 \cdot \alpha_{ey} T_e \hat{E}_y$$

$$\Gamma_{ey} = \sum_{a=0}^{n_s-1} z_a \Gamma_{ay}^{(he)}$$

$$\Gamma_{ay}^{(he)} = \left(\frac{5}{3} V_a^{(AN)} + V_{ay}^{(E)} \right) n_a - \frac{5}{3} \left[D_a^n \frac{1}{h_y} \frac{\partial n_a}{\partial y} + \tilde{D}_{ay}^p \frac{1}{h_y} \frac{\partial p_a}{\partial y} \right]$$

$$V_{ey}^{(Str)} = 0$$

$$\hat{E}_y = \frac{1}{e n_e h_y} \frac{\partial n_e T_e}{\partial y} - \frac{1}{h_y} \frac{\partial \phi}{\partial y},$$

$$\kappa_{ey} = \kappa_e^{(AN)} + \kappa_{ey}^{(CL)} + \kappa_e^{(ST)},$$

$$\kappa_{ey}^{(CL)} = 0$$

$$\kappa_e^{(ST)} = \begin{cases} 1.6\alpha^{(ST)}\sigma^{(ST)}T_e/e^2, & y \in [-\Delta^{(ST)}, 0] \\ 0, & y \notin [-\Delta^{(ST)}, 0] \end{cases}, \quad \alpha^{(ST)} \approx 2$$

$$\alpha_{ey} = \alpha_{ANy} + \alpha_{CLy},$$

$$\alpha_{ANy} = \left(c_{\alpha,0} + c_{\alpha,1} \frac{10^{20}}{n_e} + c_{\alpha,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \right) n_e \sqrt{\frac{e}{T_e}}$$

$$c_{\alpha,0} = 10^{-3}, \quad c_{\alpha,1} = 0, \quad c_{\alpha,3} = 0$$

$$\alpha_{CLy} = 0$$

$$\begin{aligned} & \frac{3}{2} \frac{\partial n_i T_i}{\partial t} + \frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} \tilde{q}_{ix} \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \right) + \sum_{a=0}^{n_s-1} \frac{n_a T_i}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} V_{a||} b_x \right) = \\ & = Q_{\Delta} + Q_{F_{ab}} + c_{E \times B} T_i B_z \frac{1}{h_x h_y} \left(\frac{\partial \phi}{\partial y} \frac{\partial}{\partial x} \left(\frac{1}{B^2} \right) - \frac{\partial \phi}{\partial x} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right) \right) \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} n_a + \\ & + \sum_{a=0}^{n_s-1} \left(\eta_{ax} \left(\frac{\partial V_{a||}}{h_x \partial x} \right)^2 + \eta_{ANa} \left(\frac{\partial V_{a||}}{h_y \partial y} \right)^2 \right) + Q_I^{(i)} + Q_R^{(i)}, \end{aligned}$$

If Eirene is working then: $\sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \left(\eta_{ax} \left(\frac{\partial V_{a||}}{h_x \partial x} \right)^2 + \eta_{ANa} \left(\frac{\partial V_{a||}}{h_y \partial y} \right)^2 \right)$

Contribution from friction between charged species

Old form

$$Q_{F_{ab}} = \sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} \sum_{\substack{a=b+1 \\ z_a \neq 0}}^{n_s-1} \frac{1}{4\tau_p} m_p \sqrt{\frac{m_a}{m_p} \frac{m_b}{m_p}} z_a^2 z_b^2 n_a n_b (V_{a||} - V_{b||})^2$$

New form is used when b2sigp_style is 2

$$Q_{F_{ab}} = - \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} V_{a||} (S_{fr,a}^m + S_{Term,a}^m)$$

Atom heat source due to ionization

$$Q_I^{(i)} = \sum_{a=0}^{n_s-2} Q_{I_a}^{(i)},$$

$$Q_{I_a}^{(i)} = \begin{cases} r_{I_a} n_e n_a \left(\frac{m_a + m_{a+1}}{4} \right) (V_{||a} - V_{||a+1})^2, & z_a < z_{a+1} \quad \& \quad z_n = z_{n+1} \\ 0 \end{cases}$$

Atom heat source due to recombination

$$Q_R^{(i)} = \sum_{a=1}^{n_s-1} Q_{R_a}^{(i)},$$

$$Q_{R_a}^{(i)} = \begin{cases} r_{R_a} n_e n_a \left(\frac{m_a + m_{a-1}}{4} \right) (V_{\parallel a} - V_{\parallel a-1})^2, & z_{a-1} < z_a \quad \& \quad z_{n-1} = z_n \\ 0 & \end{cases}$$

$$\tau_p = \frac{3}{4\sqrt{2\pi}} \frac{\sqrt{m_p} T_i^{\frac{3}{2}}}{\Lambda} \left(\frac{4\pi\epsilon_0}{e^2} \right)^2, \text{ where the Coulomb logarithm } \Lambda \text{ can be chosen as}$$

$$\Lambda = \begin{cases} 12 & \\ \begin{cases} 23,4 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 3,45 \cdot \log_{10} \left(\frac{T_e}{e} \right), & \frac{T_e}{e} \leq 50 \\ 25,3 - 1,15 \cdot \log_{10} \left(\frac{n_e}{10^6} \right) + 2,30 \cdot \log_{10} \left(\frac{T_e}{e} \right), & \frac{T_e}{e} > 50 \end{cases} & \text{Braginsky formula} \\ \max \left\{ 5; 17,3 - 0,5 \cdot \log_e \left(\frac{n_e}{10^{20}} \right) + 1,5 \cdot \log_e \left(\frac{T_e}{e \cdot 10^3} \right) \right\} & \text{Wesson formula for ion-ion collisions} \end{cases},$$

$$\tilde{q}_{ix} = \tilde{f}_{ix} T_i - \kappa_{ix} \frac{1}{h_x} \frac{\partial T_i}{\partial x} - q_{ix}^{P.Sch}$$

$$\tilde{f}_{ix} = \frac{3}{2} \Gamma_{ix} + V_{ix}^{(Str)} n_i + \frac{3}{2} \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis||)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{e}$$

$$q_{ix}^{P.Sch} = \frac{5}{2} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} D_{axy} \frac{1}{h_y} \frac{\partial n_a T_i^2}{\partial y}$$

$$q_{ix} = \left[\frac{3}{2} \Gamma_{ix} + \frac{5}{2} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} n_a \tilde{V}_{ax}^{(dia)} + V_{ix}^{(Str)} n_i + \frac{3}{2} \frac{j_x^{(AN)} + j_x^{(in)} + \tilde{j}_x^{(vis||)} + \tilde{j}_x^{(visq)} + j_x^{(vis\perp)}}{e} \right] T_i - \kappa_{ix} \frac{1}{h_x} \frac{\partial T_i}{\partial x}$$

$$\Gamma_{ix} = \sum_{a=0}^{n_s-1} \Gamma_{ax}^{(he)}$$

$$q_{ix}^{(Eir)} = \frac{5}{2} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \Gamma_{ax}^{(Eir)} T_i - \kappa_{ix} \frac{1}{h_x} \frac{\partial T_i}{\partial x}$$

$$\kappa_{ix} = \kappa_{ix}^{(AN)} + \tilde{\kappa}_{ix}^{(CL)},$$

$$k_{ix}^{(AN)} = \sum_{a=0}^{n_s-1} \tilde{k}_{ax}^{(AN)},$$

$$\tilde{k}_{ax}^{(AN)} = \begin{cases} k_a^{(AN)}, & z_a \neq 0 \\ \frac{k_a^{(AN)}}{\left(1 + \frac{\left(\frac{\sqrt{g}}{h_x} k_a^{(AN)} \left| \frac{\partial T_i}{h_x \partial x} \right| \right)^{\gamma_2}}{\frac{\sqrt{g}}{h_x} \alpha V_{h_i} n_a T_i} \right)^{\frac{1}{\gamma_2}}}, & z_a = 0 \end{cases}$$

$$V_{h_i} = \sqrt{\frac{2T_i}{m_a}}$$

$$k_a^{(AN)} = \chi_a^{(AN)} n_a,$$

$$\chi_a^{(AN)} = \left(c_{h_a,0} + c_{h_a,1} \frac{10^{20}}{n_e} + c_{h_a,2} D_a^f + c_{h_a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} \right) f_{h_a}(x,y),$$

$$a=0, \ c_{h_a,0}=0, \ c_{h_a,1}=0, \ c_{h_a,2}=1.4, \ c_{h_a,3}=0,$$

$$a=1, \ c_{h_a,0}=0, \ c_{h_a,1}=0, \ c_{h_a,2}=1.2, \ c_{h_a,3}=0,$$

$$f_{h_a}(x,y) = \begin{cases} 1, & z_a = 0 \\ \alpha_{1,h_a}, & -\delta_{1,h_a} \leq y \leq 0, \ x \in Core, \ z_a \neq 0 \\ \alpha_{2,h_a}, & 0 \leq y \leq \delta_{2,h_a}, \ x \in Core, \\ \alpha_{3,h_a}, & x,y \in Pr \cup SOL \end{cases}$$

$$\tilde{\kappa}_{ix}^{(CL)} = c_{k_{ix}}^{(F\lim)} \tilde{k}_{ix}^{(CL)}$$

$$c_{k_{ix}}^{(F\lim)} = \begin{cases} 1, & \text{for inner flux surfaces} \\ \frac{1}{1 + \frac{\frac{\sqrt{g}}{h_x} k_{ix}^{(CL)} \left| \frac{\partial T_i}{h_x \partial x} \right|}{V_{h_i}^{(\max)} n_i T_i}}, & \text{for other flux surfaces} \end{cases},$$

$$V_{h_i}^{(\max)} = c_{f\lim_1} \frac{\sqrt{g}}{h_x} |b_x| \sqrt{\frac{T_i}{m_p}} \frac{\sum_{a=0}^{n_s-1} n_a \sqrt{\frac{m_p}{m_a}}}{n_i},$$

$$c_{f\lim_1}=0,15$$

$$\tilde{\kappa}_{ix}^{(CL)} = c_{k_{ix}}^{(Luciani)} \kappa_{ix}^{(CL)},$$

'b2trcl_lluciani' '1'

$$c_{k_{ix}}^{(Luciani)} = \begin{cases} \frac{1}{7.5 \cdot 10^{16} \left(\frac{T_i}{e} \right)^2}, & \text{for inner flux surfaces} \\ 1 + \frac{c(y) \sum_{a=0}^{n_s-1} z_a^2 n_a}{1}, & \text{for other flux surfaces} \end{cases}$$

For plateau and banana correction 'b2trcl_lluciani' '3'

$$c_{k_{ix}}^{(Luciani)} = 1.6 \cdot \varepsilon^{3/2} / K_2,$$

$$K_2 = \frac{0.66 + 1.88 \cdot \sqrt{\varepsilon} - 1.54 \cdot \varepsilon}{1 + 1.03 \cdot \nu_*^{1/2} + 0.31 \cdot \nu_*} + \frac{1.17 \cdot \varepsilon^3 \cdot \nu_*}{1 + 0.74 \cdot \varepsilon^{3/2} \cdot \nu_*}$$

$$\kappa_{ix}^{(CL)} = \frac{5}{2} b_x^2 \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \frac{T_i}{m_a} n_a f_{k_i} \tau_a,$$

$$f_{k_i} = 2.253$$

For the integral modeling ('b2trcl_integral_modeling' '1') in the region of the overlap with ASTRA in the inner part of the core:

$$\tilde{\kappa}_{ix}^{(CL)} = \frac{25}{4} \left(n_1 T_i \frac{B_z}{B_x h_y} \right)^2 \frac{h_{y(\text{outermidplane})}}{e} \left[\oint \frac{\sqrt{g}}{B^2} \left(1 - \frac{B^2}{\langle B^2 \rangle} \right)^2 dx \right] \times K_{ASTRA},$$

$$\text{here } \langle B^2 \rangle = \frac{\oint B^2 \sqrt{g} dx}{\oint \sqrt{g} dx}$$

from ASTRA we get T_i/e and $\oint q_i^{NEO} dS$ as a functions of $R_{(\text{outermidplane})}$ and calculate

$$K_{ASTRA} = \frac{-1}{\oint q_i^{NEO} dS} \frac{d(T_i/e)}{dR_{(\text{outermidplane})}} \text{ as a function of the flux surface in B2}$$

$$V_{ix}^{(Str)} = 0$$

$$\tilde{q}_{iy} = \tilde{f}_{iy} T_i - \kappa_{iy} \frac{1}{h_y} \frac{\partial T_i}{\partial y} - q_{iy}^{P.Sch}$$

$$\tilde{f}_{iy} = \frac{3}{2} \Gamma_{iy} + V_{iy}^{(Str)} n_i + \frac{3}{2} \frac{(j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + \tilde{j}_y^{(visq)} + j_y^{(vis\perp)})}{e}$$

$$q_{iy}^{P.Sch} = -\frac{5}{2} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i^2}{\partial x}$$

$$q_{iy} = \left[\frac{3}{2} \Gamma_{iy} + \frac{5}{2} \sum_{a=0}^{n_s-1} n_a \tilde{V}_{ay}^{(dia)} + V_{iy}^{(Str)} n_i + \frac{3}{2} \beta_{\text{Scott}} \frac{j_y^{(in)}}{e} + \frac{3}{2} \frac{\left(j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + \tilde{j}_y^{(visq)} + j_y^{(vis\perp)} \right)}{e} \right] T_i - \kappa_{iy} \frac{1}{h_y} \frac{\partial T_i}{\partial y}$$

$$\beta_{\text{Scott}} = \begin{cases} 0, & \text{for us} \\ 1, & \text{for Bruce Scott} \end{cases}$$

$$\Gamma_{iy} = \sum_{a=0}^{n_s-1} \Gamma_{ay}^{(he)}$$

$$q_{iy}^{(Eir)} = \frac{5}{2} \sum_{a=0}^{n_s-1} \Gamma_{ay}^{(Eir)} T_i - \kappa_{iy} \frac{1}{h_y} \frac{\partial T_i}{\partial y}$$

$$V_{iy}^{(Str)} = 0$$

$$\kappa_{iy} = \kappa_{iy}^{(AN)} + \kappa_{iy}^{(CL)},$$

$$k_{iy}^{(AN)} = \sum_{a=0}^{n_s-1} \tilde{k}_{ay}^{(AN)},$$

$$\tilde{k}_{ay}^{(AN)} = \begin{cases} k_a^{(AN)}, & z_a \neq 0 \\ \frac{k_a^{(AN)}}{\left(1 + \frac{\left(\frac{\sqrt{g}}{h_y} k_a^{(AN)} \left| \frac{\partial T_i}{h_y \partial y} \right| \right)^{\gamma_2}}{\frac{\sqrt{g}}{h_y} \alpha V_{h_i} n_a T_i} \right)^{\frac{1}{\gamma_2}}}, & z_a = 0 \end{cases}$$

$$\kappa_{iy}^{(CL)} = 0$$

$$\frac{1}{\sqrt{g}} \frac{\partial}{\partial x} \left(\frac{\sqrt{g}}{h_x} j_x \right) + \frac{1}{\sqrt{g}} \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} j_y \right) = 0$$

$$j_x = j_x^{(AN)} + \tilde{j}_x^{(dia)} + j_x^{(in)} + \tilde{j}_x^{(vis||)} + \tilde{j}_x^{(vis\perp)} + \tilde{j}_x^{(visq)} + \tilde{j}_x^{(s)} + j_x^{(ll)},$$

$$j_x^{(AN)} = -\sigma_{AN} \frac{\partial \phi}{h_x \partial x}$$

$$\tilde{j}_x^{(dia)} = B_z \frac{\sum_{a=0}^{n_s-1} n_a (z_a T_e + T_i)}{h_y} \frac{\partial}{\partial y} \left(\frac{1}{B^2} \right),$$

$$j_x^{(in)} = \frac{B_z}{2} \frac{\partial}{h_y \partial y} \left(\frac{1}{B^2} \right) \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} m_a n_a V_{a||}^2,$$

$$\tilde{j}_x^{(vis||)} = -\frac{b_z B_x}{3B^{\frac{1}{2}}} \frac{\partial}{h_y \partial y} \left(\frac{1}{B^2} \right) \sum_{\substack{a=0 \\ z_{n_a}=1 \text{ \& } z_a=1}}^{n_s-1} \tilde{\eta}_{CLax} \frac{\partial(\sqrt{BV_{a||}})}{h_x \partial x},$$

$$\tilde{j}_x^{(vis\perp)} = 0,$$

$$\tilde{j}_x^{(visq)} = -0.24 \sum_{\substack{a=0 \\ z_{n_a}=1 \text{ \& } z_a=1}}^{n_s-1} \alpha^{NEO} \tilde{\tau}_a b_z \frac{B_x}{B^{\frac{1}{2}}} \frac{\partial}{h_y \partial y} \left(\frac{1}{B^2} \right) \frac{\partial \left(\tilde{q}_a^{(0)} B^{\frac{1}{2}} \right)}{h_x \partial x}, \quad \tilde{q}_a^{(0)} \text{ see above}$$

$$\tilde{j}_x^{(s)} = \sum_{\substack{a_0=0 \\ z_{a_0}=0 \text{ \& } z_{n_{a_0}}=1}}^{n_s-1} \sum_{\substack{a=0 \\ z_a \neq 0 \text{ \& } z_{n_a}=1}}^{n_s-1} \left(-\sigma_{aa_0} b_z^2 \frac{\partial \phi}{h_x \partial x} - \sigma_{aa_0} b_z^2 \frac{1}{Z_a e n_a} \frac{\partial_a T_i}{h_x \partial x} + \sigma_{aa_0} B_z V_{a_0 y} \right),$$

$$V_{a_0 y} = \frac{\Gamma_{a_0 y}}{n_{a_0}}$$

$$\sigma_{aa_0} = \begin{cases} \frac{m_a n_a \langle V_{aa_0} \sigma_{ex} \rangle n_{a_0}}{2B^2}, & z_{a_0} = 0 \text{ \& } z_{n_{a_0}} = 1, \quad z_a \neq 0 \text{ \& } z_{n_a} = 1, \\ 0 & \end{cases}$$

$$\langle V_{aa_0} \sigma_{ex} \rangle = 3.2 \cdot 10^{-15} \sqrt{\frac{T_i}{0.026e}}$$

$$j_x^{(||)} - \text{see above}$$

$$j_y = j_y^{(AN)} + \tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)} + j_y^{(ST)},$$

$$j_y^{(AN)} = -\sigma_{AN} \frac{\partial \phi}{h_y \partial y},$$

$$\tilde{j}_y^{(dia)} = -\frac{B_z \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} n_a (z_a T_e + T_i)}{h_x} \frac{\partial}{\partial x} \left(\frac{1}{B^2} \right),$$

$$j_y^{(in)} = -\frac{B_z}{2} \frac{\partial}{h_x \partial x} \left(\frac{1}{B^2} \right) \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} m_a n_a V_{a||}^2,$$

$$j_y^{(vis||)} = b_z \frac{B_x}{3B^{\frac{1}{2}}} \frac{\partial}{h_x \partial x} \left(\frac{1}{B^2} \right) \sum_{\substack{a=0 \\ z_{n_a}=1 \& z_a=1}}^{n_s-1} \tilde{\eta}_{CLax} \frac{\partial \left(B^{\frac{1}{2}} V_{a||} \right)}{h_x \partial x}, \text{ not agreed with parallel momentum balance?}$$

$$j_y^{(vis\perp)} = - \sum_{\substack{a=0 \\ z_{n_a}=1 \& z_a=1}}^{n_s-1} \frac{1}{B\sqrt{g}} \frac{\partial}{\partial y} \left(\frac{\sqrt{g}}{h_y} \frac{\eta_{ANa}}{4} \frac{\partial \mathcal{N}_{a\perp}}{h_y \partial y} \right),$$

$$V_{a\perp} = - \frac{1}{Bh_y} \frac{\partial \phi}{\partial y} - \frac{1}{z_a e B h_y n_a} \frac{\partial n_a T_i}{\partial y}$$

$$j_y^{(visq)} = 0.24 \sum_{\substack{a=0 \\ z_{n_a}=1 \& z_a=1}}^{n_s-1} \alpha^{NEO} \tilde{\tau}_a b_z \frac{B_x}{B^2} \frac{\partial}{h_x \partial x} \left(\frac{1}{B^2} \right) \frac{\partial \left(\tilde{q}_a^{(0)} B^{\frac{1}{2}} \right)}{h_x \partial x}, \quad \tilde{q}_a^{(0)} \text{ see above}$$

$$j_y^{(s)} = \sum_{\substack{a_0=0 \\ z_{a_0}=0 \& z_{n_{a_0}}=1}}^{n_s-1} \sum_{\substack{a=0 \\ z_a \neq 0 \& z_{n_a}=1}}^{n_s-1} \left(-\sigma_{aa_0} \frac{\partial \phi}{h_y \partial y} - \sigma_{aa_0} \frac{1}{z_a e n_a} \frac{\partial n_a T_i}{h_y \partial y} - \sigma_{aa_0} B_z V_{a_0 x} \right),$$

$$V_{a_0 x} = \frac{\Gamma_{a_0 x}}{n_{a_0}}$$

$$\sigma_{aa_0} \text{ see above}$$

$$j_y^{(ST)} = \begin{cases} -\sigma^{(ST)} \left(\frac{1}{h_y} \frac{\partial \phi}{\partial y} - \frac{T_e}{e n_e} \frac{1}{h_y} \frac{\partial n_e}{\partial y} - \frac{0.5}{e} \frac{1}{h_y} \frac{\partial T_e}{\partial y} \right), & y \in [-\Delta^{(ST)}, 0] \\ 0, & y \notin [-\Delta^{(ST)}, 0] \end{cases}$$

$$\sigma^{(ST)} = \begin{cases} c_{\sigma^{(ST)}} e n_e, & y \in [-\Delta^{(ST)}, 0] \\ 0, & y \notin [-\Delta^{(ST)}, 0] \end{cases}, \quad c_{\sigma^{(ST)}} > c_{\sigma_{ANy}}$$

$$\sigma_y = \sigma_{AN} + \sigma_{CLy}$$

$$\sigma_{AN} \text{ depends on 'b2tqna_model_sig'}$$

$$\sigma_{AN} = \left(c_{\sigma_a,0} + c_{\sigma_a,1} \frac{10^{20}}{n_e} + c_{\sigma_a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} + c_{\sigma_a,4} e^{-c_{\sigma_a,5} \left(\frac{y}{L(x)} \right)^2} \right) e n_e,$$

or

$$\sigma_{AN} = \left(c_{\sigma_a,0} + c_{\sigma_a,1} \frac{10^{20}}{n_e} + c_{\sigma_a,3} \frac{6,2 \cdot 10^{-2} T_e}{e|B|} + c_{\sigma_a,4} e^{-c_{\sigma_a,5} \left(\frac{y}{L(x)} \right)^2} \right) e \tilde{n}_e, \quad \tilde{n}_e - \text{density at inner flux surface}$$

$$L(x) = \int_0^{y_{out}} h_y(x, y) dy$$

$$c_{\sigma_a,0} = 10^{-6} - 10^{-3}, \quad c_{\sigma_a,1} = 0, \quad c_{\sigma_a,3} = 0,$$

$$\sigma_{CLy} = 0$$

Typical boundary conditions

i. Particle balance equation

$$S_{n_a} = S_{n_a,0} + S_{n_a,1} n_a$$

South. Private region

$$z_a = 0$$

$$\left. \frac{\sqrt{g}}{h_y} \Gamma_{ay} \right|_{S_0} = \left[p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a \right]_{S_0} + \sum_{a=0}^{n_s-1} r_{rec,a,0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \Bigg]_{S_0} \text{ leakage, BCCON=10}$$

$$p_{n_a,a,1} = -10^{-2}, r_{rec,0,0} = 0, r_{rec,1,0} = 1$$

$\tilde{\Gamma}_{ay}$ was replaced by Γ_{ay}

c_{nsr} -neutral sources rescale coefficient. 1 – when working B2 and 10^{-10} when eirene coupling

$$z_a \neq 0$$

BCCON=9 is below

$$\left. \frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \right|_{S_0} = \begin{cases} \left[-\frac{1}{p_{n_a,a,1}} \frac{\sqrt{g}}{h_y} (D_{ANa}^n + D_{ANa}^p (z_a T_e + T_i)) n_a \right]_{S_0}, & mdf_fnb = 0 \\ \left[-\frac{1}{p_{n_a,a,1}} \frac{\sqrt{g}}{h_y} (D_{ANa}^n + D_{ANa}^p (z_a T_e + T_i)) n_a + \frac{\sqrt{g}}{h_y} D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i}{\partial x} + \frac{\sqrt{g}}{h_y} V_{ay}^{(E)} n_a \right]_{S_0}, & mdf_fnb = 1 \end{cases}$$

$$S_{n_a,0}^{no_mdf} = 0$$

$$S_{n_a,1}^{no_mdf} = -\frac{1}{p_{n_a,a,1}} \frac{\sqrt{g}}{h_y} (D_{ANa}^n + D_{ANa}^p (z_a T_e + T_i))$$

BCCON=10 (leakage) is below

$$\left. \frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \right|_{S_0} = \begin{cases} \left[p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a + \min \left\{ 0, \frac{\sqrt{g}}{h_y} n_a V_{ay}^{(E)} \right\} \right]_{S_0}, & mdf_fnb = 0 \\ \left[p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a + \frac{\sqrt{g}}{h_y} D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i}{\partial x} + \min \left\{ 0, \frac{\sqrt{g}}{h_y} n_a V_{ay}^{(E)} \right\} \right]_{S_0}, & mdf_fnb = 1 \end{cases}$$

$$p_{n_a,a,1} = -10^{-3}$$

$$S_{n_a,0}^{no_mdf} = 0$$

$$S_{n_a,1}^{no_mdf} = p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a$$

BCCON=15 is below

$$-\frac{\sqrt{g}}{h_y^2} D_{ay}^n \frac{\partial n_a}{\partial y} \Big|_{S_0} = \left[c_{n_a,a,2} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a \right] \Big|_{S_0}$$

$$c_{n_a,a,2} = -10^{-3}$$

South 1. Core plasma

$$z_a = 0$$

$$\frac{\sqrt{g}}{h_y} \Gamma_{ay} \Big|_{S_1} = 0 \quad \text{total particle flux (8), dpc}$$

$$\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \Big|_{S_1} = 0$$

$$z_a \neq 0$$

constant density (1)

$$\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \Big|_{S_1} = \frac{\sqrt{g}}{h_y} (v_{big} n_a \Big|_{S_1} - v_{big} n_a) \Big|_{S_1}, \quad v_{big} \approx 10^{12}, \quad n_a \Big|_{S_1} \text{ is a given density}$$

the total particle flux with constant flux density (8)

$$\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \Big|_{S_1} = p_{n,a,0} \frac{\frac{\sqrt{g}}{h_y}}{\sum_i \left(\frac{\sqrt{g}}{h_y} \right)_{i,-1} \Big|_{S_1}} + \frac{\sqrt{g}}{h_y} D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i}{\partial x} + \frac{\sqrt{g}}{h_y} V_{ya}^{(E)} n_a \Big|_{S_1}$$

North

$$z_a = 0$$

$$-\frac{\sqrt{g}}{h_y}\Gamma_{ay}\Big|_N = \left[p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a \right]_N + \sum_{a=0}^{n_s-1} r_{rec,a,0} c_{nsr} \max \left\{ 0, \frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \Big|_N \quad \text{leakage (10), dpc}$$

$$p_{n_a,a,1} = 0, r_{rec,0,0} = 0, r_{rec,1,0} = 1$$

$$-\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\Big|_N = \sum_{a=0}^{n_s-1} c_{rec,a,0} c_{nsr} \max \left\{ 0, \frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \quad \tilde{\Gamma}_{ay} \text{ was replaced by } \Gamma_{ay}$$

$$c_{rec,0,0} = 0, c_{rec,1,0} = 1$$

$$z_a \neq 0$$

BCCON=9 is below

$$-\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\Big|_N = \begin{cases} \left[\frac{1}{p_{n_a,a,1}} \frac{\sqrt{g}}{h_y} (D_{ANa}^n + D_{ANa}^P (z_a T_e + T_i)) n_a \right]_{S_0}, & mdf_fnb = 0 \\ \left[\frac{1}{p_{n_a,a,1}} \frac{\sqrt{g}}{h_y} (D_{ANa}^n + D_{ANa}^P (z_a T_e + T_i)) n_a - \frac{\sqrt{g}}{h_y} D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i}{\partial x} - \frac{\sqrt{g}}{h_y} V_{ay}^{(E)} n_a \right]_N, & mdf_fnb = 1 \end{cases}$$

$$S_{n_a,0}^{no_mdf} = 0$$

$$S_{n_a,1}^{no_mdf} = \frac{1}{p_{n_a,a,1}} \frac{\sqrt{g}}{h_y} (D_{ANa}^n + D_{ANa}^P (z_a T_e + T_i))$$

BCCON=10 (leakage) is below

$$-\frac{\sqrt{g}}{h_y}\tilde{\Gamma}_{ay}\Big|_N = \begin{cases} \left[p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a - \max \left\{ 0, \frac{\sqrt{g}}{h_y} n_a V_{ay}^{(E)} \right\} \right]_N, & mdf_fnb = 0 \\ \left[p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a - \frac{\sqrt{g}}{h_y} D_{ayx} \frac{1}{h_x} \frac{\partial n_a T_i}{\partial x} - \max \left\{ 0, \frac{\sqrt{g}}{h_y} n_a V_{ay}^{(E)} \right\} \right]_N, & mdf_fnb = 1 \end{cases}$$

$$S_{n_a,0}^{no_mdf} = 0$$

$$S_{n_a,1}^{no_mdf} = p_{n_a,a,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a$$

$$\frac{\sqrt{g}}{h_y^2} D_{ay}^n \frac{\partial n_a}{\partial y} \Big|_N = \left[c_{n_a, a, 2} \frac{\sqrt{g}}{h_y} \sqrt{\frac{z_a T_e + T_i}{m_a}} n_a \right]_N \quad \text{BCCON=15}$$

$$c_{n_a, a, 2} = -10^{-3}$$

North with puffing

$$z_a = 0$$

$$-\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \Big|_N = p_{n, a, 0} \frac{\sqrt{g}}{h_y} \Big|_N + \sum_{a=0}^{n_s-1} \left[c_{rec, a, 0} c_{nsr} \max \left\{ 0, \frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \right] \Big|_N \quad \text{the particle flux per unit area (5)}$$

or $\tilde{\Gamma}_{ay}$ was replaced by Γ_{ay}

$$-\frac{\sqrt{g}}{h_y} \tilde{\Gamma}_{ay} \Big|_N = \frac{p_{n, a, 0}}{\sum_{i=0}^{N_x-1} \left(\frac{\sqrt{g}}{h_y} \right)_{i, N_y}} \frac{\sqrt{g}}{h_y} \Big|_N + \sum_{a=0}^{n_s-1} \left[c_{rec, a, 0} c_{nsr} \max \left\{ 0, \frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \right] \Big|_N \quad \text{the total particle flux with constant}$$

flux density (8)

$\tilde{\Gamma}_{ay}$ was replaced by Γ_{ay}

$$c_{rec, 0, 0} = 0, c_{rec, 1, 0} = 1, p_{n, 0a, 0} \sim 10^{21},$$

West

$$z_a = 0$$

$$\frac{\sqrt{g}}{h_x} \Gamma_{ax} \Big|_W = \left[p_{n, a, 1} \frac{\sqrt{g}}{h_x} + \sum_{a=0}^{n_s-1} r_{rec, a, 0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_x} \Gamma_{ax} \right\} \right] \Big|_W \quad \text{particle flux per area (5), dpc}$$

$$p_{n_a, a, 1} = 0, r_{rec, 0, 0} = 0, r_{rec, 1, 0} = 1$$

$$\frac{\sqrt{g}}{h_x} \tilde{\Gamma}_{ax} \Big|_W = \left[p_{n, a, 1} \frac{\sqrt{g}}{h_x} + \sum_{a=0}^{n_s-1} c_{rec, a, 0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_x} \Gamma_{ax} \right\} \right] \Big|_W \quad \text{It has been replaced } \tilde{\Gamma}_{ax} \text{ by } \Gamma_{ax}. \quad (5)$$

$$p_{n_a, a, 1} = 0, c_{rec, 0, 0} = 0, c_{rec, 1, 0} = 1$$

$$z_a \neq 0$$

$$\left. \frac{\sqrt{g}}{h_x} \Gamma_{ax} \right|_W = 10^{20} \left. \frac{\sqrt{g}}{h_x} \frac{\partial n_a}{\partial x} \right|_W \quad \text{sheath density bc (3) dpc}$$

$$\left. \frac{\sqrt{g}}{h_x} \tilde{\Gamma}_{ax} \right|_W = \left[-p_{n,a,1} \frac{\sqrt{g}}{h_x} \left| b_x \right| \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} m_b n_b}} n_a - \frac{\sqrt{g}}{h_x} D_{axy} \frac{1}{h_y} \frac{\partial n_a T_i}{\partial y} \right]_W \quad \text{sound speed flux bc (14)}$$

$$p_{n,a,1} = +1,$$

East

$$z_a = 0$$

$$\left. -\frac{\sqrt{g}}{h_x} \Gamma_{ax} \right|_E = \left[p_{n,a,1} \frac{\sqrt{g}}{h_x} + \sum_{a=0}^{n_s-1} r_{rec,a,0} c_{nsr} \max \left\{ 0, \frac{\sqrt{g}}{h_x} \Gamma_{ax} \right\} \right]_E \quad \text{particle flux per area (5), dpc}$$

$$p_{n_a,a,1} = 0, r_{rec,0,0} = 0, r_{rec,1,0} = 1$$

$$\left. -\frac{\sqrt{g}}{h_x} \tilde{\Gamma}_{ax} \right|_E = \left[\sum_{a=0}^{n_s-1} c_{rec,a,0} c_{nsr} \max \left\{ 0, \frac{\sqrt{g}}{h_x} \Gamma_{ax} \right\} \right]_E \quad \text{It has been replaced } \tilde{\Gamma}_{ax} \text{ by } \Gamma_{ax}. (5)$$

$$c_{rec,0,0} = 0, c_{rec,1,0} = 1,$$

$$z_a \neq 0$$

$$\left. -\frac{\sqrt{g}}{h_x} \Gamma_{ax} \right|_E = -10^{20} \left. \frac{\sqrt{g}}{h_x} \frac{\partial n_a}{\partial x} \right|_E \quad \text{sheath density bc (3) dpc}$$

$$\left. -\frac{\sqrt{g}}{h_x} \tilde{\Gamma}_{ax} \right|_E = \left[-p_{n,a,1} \frac{\sqrt{g}}{h_x} \left| b_x \right| \sqrt{\frac{\sum_{b=0, z_b \neq 0}^{n_s-1} p_b}{\sum_{b=0, z_b \neq 0}^{n_s-1} m_b n_b}} n_a + \frac{\sqrt{g}}{h_x} D_{axy} \frac{1}{h_y} \frac{\partial n_a T_i}{\partial y} \right]_E \quad \text{sound speed flux bc (14)}$$

$$p_{n,a,1} = +1$$

ii. **Parallel momentum balance for ions**

$$S_{m_a} = S_{m_a,0} + S_{m_a,1} u_{\parallel a} + S_{m_a,2} m_a n_a + S_{m_a,3} m_a n_a u_{\parallel a}$$

South. Private region

$$z_a = 0,$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\parallel a} \right] \Big|_{S_0} \quad \text{weak velocity (4) dpc}$$

$$p_{m,a,1} = 0, \quad p_{m,a,2} = 10^5$$

We had this b.c.

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} + c_{m,a,6} \max \left(0, -h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\parallel a} \right] \Big|_{S_0}, \quad \text{delete term with max !}$$

$$\Gamma_{ay}^{(cor)} = \Gamma_{ay} + \Gamma_{ay}^{(dia)}$$

$$c_{m,a,3} = -10^{10}, \text{ we have } c_{m,a,6} = -1, \text{ but not } c_{m,a,6} = 0$$

Inserted into b2stbc_phys for bcmom (12) as $c_{m,a,3} \rightarrow p_{m,a,2} = -10^{10}, c_{m,a,6} \rightarrow p_{m,a,1} = -1$

$$z_a \neq 0,$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\parallel a} \right] \Big|_{S_0} \quad \text{weak velocity (4) dpc}$$

$$p_{m,a,1} = 0, \quad p_{m,a,2} = 10^5$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_0} = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} + c_{m,a,6} \max \left(0, -h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\parallel a} \right] \Big|_{S_0}, \quad \text{delete term with max !}$$

$$\Gamma_{ay}^{(cor)} = \Gamma_{ay} + \Gamma_{ay}^{(dia)}$$

$$c_{m,a,3} = 0, \quad c_{m,a,6} = -1$$

South 1. Core

$$z_a = 0$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\parallel a} \right] \Big|_{S_1} \quad \text{weak velocity (4) dpc}$$

$$p_{m,a,1} = 0, \quad p_{m,a,2} = 10^5$$

$$\text{BCMOM}=12 \quad p_{m,a,2} = -10^{10}, \quad p_{m,a,1} = 0$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} \Big|_{S_1}$$

$$z_a \neq 0$$

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\parallel a} \right] \Big|_{S_1} \quad \text{weak velocity (4) dpc}$$

$$p_{m,a,1} = 0, \quad p_{m,a,2} = 10^5$$

$$\text{BCMOM}=14 \quad p_{m,a,1} = 0, \quad p_{m,a,3} = -10^{10}$$

$$\begin{aligned} h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} &= p_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} \Big|_{S_1} + \left[h_z \frac{\sqrt{g}}{h_y} m_a \langle n_a \rangle (-2) \frac{\langle T_i \rangle B_z}{e z_a} \frac{\partial}{h_x \partial x} \left(\frac{1}{B^2} \right) \langle V_{\parallel a} \rangle c_{drift} + c_{corflux} h_z \frac{\sqrt{g}}{h_y} \right] \Big|_{S_1} + \\ &+ p_{m,a,1} \max \left(0, -h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\parallel a} \Big|_{S_1} \end{aligned}$$

The value of the parallel velocity as a function of poloidal coordinate

$$h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_{S_1} = v_{big} n_{e0} m_a h_z \frac{\sqrt{g}}{h_y} \left[\frac{\langle B \rangle}{B} p_{m,a,1} - V_{\parallel a} \right] \Big|_{S_1}, \quad \text{BCMOM}=15$$

where $\langle B \rangle = \frac{\oint B \sqrt{g} dx}{\oint \sqrt{g} dx}$, the integration is done over inner magnetic surface

$v_{big} \approx 10^{12}$, $n_{e0} = 10^{20}$, $p_{m,a,1}$ is a given value of the average parallel velocity

North

$$z_a = 0,$$

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\parallel a} \right] \Big|_N \quad \text{weak velocity (4) dpc}$$

$$p_{m,a,1} = 0, \quad p_{m,a,2} = 10^5$$

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} + c_{m,a,6} \max \left(0, h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\parallel a} \right] \Big|_N \quad \text{delete term with max !}$$

$$c_{m,a,3} = -10^{10}, \text{ we have } c_{m,a,6} = -1, \text{ but not } c_{m,a,6} = 0$$

Inserted into b2stbc_phys for BCMOM=12 as $c_{m,a,3} \rightarrow p_{m,a,2} = -10^{10}, c_{m,a,6} \rightarrow p_{m,a,1} = -1$

$$z_a \neq 0,$$

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[p_{m,a,1} p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} - p_{m,a,2} h_z \frac{\sqrt{g}}{h_y} V_{\parallel a} \right] \Big|_N \quad \text{weak velocity (4) dpc}$$

$$p_{m,a,1} = 0, \quad p_{m,a,2} = 10^5$$

$$-h_z \frac{\sqrt{g}}{h_y} \Gamma_{ay}^m \Big|_N = \left[c_{m,a,3} h_z \frac{\sqrt{g}}{h_y} m_a n_a V_{\parallel a} + c_{m,a,6} \max \left(0, h_z \frac{\sqrt{g}}{h_y} \frac{\Gamma_{ay}^{(cor)}}{n_a} \right) m_a n_a V_{\parallel a} \right] \Big|_N$$

$$c_{m,a,3} = 0, \quad c_{m,a,6} = -1$$

West

$$z_a = 0, \quad p_{m,a,1} = 0$$

$$z_a \neq 0, \quad p_{m,a,1} = 1$$

$$h_z \frac{\sqrt{g}}{h_x} \Gamma_{ax}^m \Big|_W = \left[p_{m,a,1} \frac{u_{big}}{\mathcal{E}^2} V_{West} m_a n_a - \frac{u_{big}}{\mathcal{E}^2} h_z \frac{\sqrt{g}}{h_x} |b_x| m_a n_a V_{\parallel a} \right] \Big|_W \quad \text{BCMOM=13}$$

$$V_{West} = -h_z \frac{\sqrt{g}}{h_x} \frac{|b_x|}{b_x} V_{ax}^{(E \times B)} - h_z \frac{\sqrt{g}}{h_x} b_x \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}}$$

$z_a = 0$ specified parallel velocity (1) dpc

$$\frac{\sqrt{g}}{h_x} \Gamma_{ax}^m|_W = \left[u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} p_{m,a,1} - u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} V_{||a} \right]_W, \quad p_{m,a,1} = 0, \quad u_{big} = 10^{18}, \quad n_{e00} = 10^{20} \text{ dpc}$$

$z_a \neq 0$, sheath velocity (3) dpc

$p_{m,a,2} < 0.5$, sound_mode 1=0,

$$\frac{\sqrt{g}}{h_x} \Gamma_{ax}^m|_W = \left[-\frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} u_{p_{out}} - \frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} V_{||a} \right]_W, \quad \text{dpc}$$

$p_{m,a,1} = 1$, $p_{m,a,2} = 0$, $u_{big} = 10^{18}$, $n_{e00} = 10^{20}$, $\varepsilon = 10^{-10}$

$$u_{p_{out}} = \max \left\{ 0, \frac{u_{u_{out}}}{b_x} + \frac{u_a^{(dia)}}{b_x} \right\}, \quad u_{u_{out}} = p_{m,a,1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}},$$

East

$z_a = 0$, $p_{m,a,1} = 0$

$z_a \neq 0$, $p_{m,a,1} = 1$

$$-h_z \frac{\sqrt{g}}{h_x} \Gamma_{ax}^m|_E = \left[p_{m,a,1} \frac{u_{big}}{\varepsilon^2} V_{East} m_a n_a - \frac{u_{big}}{\varepsilon^2} h_z \frac{\sqrt{g}}{h_x} |b_x| m_a n_a V_{||a} \right]_E \quad \text{BCMOM=13}$$

$$V_{East} = -h_z \frac{\sqrt{g}}{h_x} \frac{|b_x|}{b_x} V_{ax}^{(E \times B)} + h_z \frac{\sqrt{g}}{h_x} b_x \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}}$$

$z_a = 0$, specified parallel velocity (1) dpc

$$-\frac{\sqrt{g}}{h_x} \Gamma_{ax}^m \Big|_E = \left[u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} p_{m,a,1} - u_{big} n_{e00} a_{m,a} \frac{\sqrt{g}}{h_x} V_{||a} \right] \Big|_E, \quad p_{m,a,1} = 0, \quad u_{big} = 10^{18}, \quad n_{e00} = 10^{20} \text{ dpc}$$

$z_a \neq 0$, sheath velocity (3) dpc

$p_{m,a,2} < 0.5$, sound_mode l=0,

$$-\frac{\sqrt{g}}{h_x} \Gamma_{ax}^m \Big|_E = \left[-\frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} u_{p_{out}} - \frac{u_{big}}{\varepsilon^2} n_a m_a \frac{\sqrt{g}}{h_x} V_{||a} \right] \Big|_E, \text{ dpc}$$

$p_{m,a,1} = 1$, $p_{m,a,2} = 0$, $u_{big} = 10^{18}$, $n_{e00} = 10^{20}$, $\varepsilon = 10^{-10}$

$$u_{p_{out}} = \max \left\{ 0, \frac{u_{out}}{b_x} - \frac{u_a^{(dia)}}{b_x} \right\}, \quad u_{out} = p_{m,a,1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}},$$

iii. Current continuity equation

$$S_{ch} = S_{ch,0} + S_{ch,1} \phi + S_{ch,2} n_e + S_{ch,3} n_e \phi$$

South. Private region

$$\frac{\sqrt{g}}{h_y} j_y \Big|_{S_0} = 0, \text{ dpc}$$

$$\left. \frac{\sqrt{g}}{h_y} j_y \right|_{S_0} = \begin{cases} 0, & \text{facdrift} = 0 \\ \left. \frac{\sqrt{g}}{h_y} \left(\tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)} \right) \right|_{S_0}, & \text{facdrift} \neq 0 \end{cases}$$

South 1. Core

$$\left. \frac{\sqrt{g}}{h_y} j_y \right|_{S_1} = \begin{cases} 0, & \text{facdrift} = 0 \\ \left. \frac{\sqrt{g}}{h_y} \left(\langle \tilde{j}_y^{(dia)} \rangle \right) \right|_{S_0}, & \text{facdrift} \neq 0 \quad \langle \tilde{j}_y^{(dia)} \rangle ?, \text{dpc} \end{cases}$$

$$\left. \frac{\sqrt{g}}{h_y} j_y \right|_{S_1} = \begin{cases} 0, & \text{facdrift} = 0 \\ \left. \frac{\sqrt{g}}{h_y} \left(\langle \tilde{j}_y^{(dia)} \rangle + \langle j_y^{(in)} \rangle \right) \right|_{S_0}, & \text{facdrift} \neq 0 \end{cases}$$

$$\langle \tilde{j}_y^{(dia)} \rangle = - \frac{B_z \left\langle \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} n_a (z_a T_e + T_i) \right\rangle}{h_x} \frac{\partial}{\partial x} \left(\frac{1}{B^2} \right),$$

$$\langle j_y^{(in)} \rangle = - \frac{B_z}{2} \frac{\partial}{h_x \partial x} \left(\frac{1}{B^2} \right) \sum_{\substack{a=1 \\ z_a \neq 0}}^{n_s-1} m_a n_a \langle V_{a||}^2 \rangle,$$

North

$$\left. \frac{\sqrt{g}}{h_y} j_y \right|_N = \begin{cases} 0, & \text{facdrift} = 0 \\ \left. \frac{\sqrt{g}}{h_y} \left(\langle \tilde{j}_y^{(dia)} \rangle \right) \right|_N, & \text{facdrift} \neq 0 \quad \langle \tilde{j}_y^{(dia)} \rangle ?, \text{dpc} \end{cases}$$

$$-\left. \frac{\sqrt{g}}{h_y} j_y \right|_N = \begin{cases} 0, & \text{facdrift} = 0 \\ -\left. \frac{\sqrt{g}}{h_y} \left(\tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)} \right) \right|_N, & \text{facdrift} \neq 0 \end{cases}$$

West

BCPOT=11 is connected with BCCON=14 when istyle_fchi = 1

$$\left. \frac{\sqrt{g}}{h_x} j_x \right|_W = \frac{\sqrt{g}}{h_x} e \left[- \sum_{a=0}^{n_s-1} z_a \left(p_{n,a,1} |b_x| \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}} n_a \right) + (1-\gamma_e) |b_x| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\phi}{T_e}\right) \right], \text{ it is new form}$$

BCPOT=11 when istyle_fchi = 0

$$\left. \frac{\sqrt{g}}{h_x} j_x \right|_W = \frac{\sqrt{g}}{h_x} e \left[\sum_{a=0}^{n_s-1} z_a \Gamma_{ax} + (1-\gamma_e) |b_x| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\phi}{T_e}\right) \right], \text{ it is old form}$$

sheath potential bc (3)

$$\left. \frac{\sqrt{g}}{h_x} j_x \right|_W = \frac{\sqrt{g}}{h_x} e \left[\sum_{a=0}^{n_s-1} z_a \Gamma_{ax} + (1-\gamma_e) |b_x| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e(\phi - p_{\phi,2})}{T_e}\right) \right], \text{ dpc}$$

East

BCPOT=11 is connected with BCCON=14 when istyle_fchi = 1

$$\left. -\frac{\sqrt{g}}{h_x} j_x \right|_E = \frac{\sqrt{g}}{h_x} e \left[- \sum_{a=0}^{n_s-1} z_a \left(p_{n,a,1} |b_x| \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}} n_a \right) + (1-\gamma_e) |b_x| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\phi}{T_e}\right) \right], \text{ it is new form}$$

BCPOT=11 when istyle_fchi = 0

$$\left. -\frac{\sqrt{g}}{h_x} j_x \right|_E = \frac{\sqrt{g}}{h_x} e \left[- \sum_{a=0}^{n_s-1} z_a \Gamma_{ax} + (1-\gamma_e) |b_x| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\phi}{T_e}\right) \right], \text{ it is old form}$$

$$c_{ch,7} = 1 - \gamma_e, \quad c_{ch,6} = 1$$

sheath potential bc (3)

$$\left. -\frac{\sqrt{g}}{h_x} j_x \right|_E = \frac{\sqrt{g}}{h_x} e \left[- \sum_{a=0}^{n_s-1} z_a \Gamma_{ax} + (1-\gamma_e) |b_x| n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e(\phi - p_{\phi,2})}{T_e}\right) \right], \text{ dpc}$$

iv. Energy balance equation for electrons

$$S_{h_e} = S_{h_e,0} + S_{h_e,1} T_e + S_{h_e,2} n_e + S_{h_e,3} n_e T_e$$

South. Private region

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \Big|_{S_0} = \left[\left(c_{h_e,2} \frac{\sqrt{g}}{h_y} \sqrt{\frac{T_e}{m_e}} \right) T_e n_e + c_{h_e,7} \max \left(0, -\frac{\sqrt{g}}{h_y} \tilde{f}_{ey} \right) T_e - q_{ex}^{P.Sch} \right] \Big|_{S_0}, \quad \text{bcene}=11$$

$$c_{h_e,2} = -10^{-4}, \quad c_{h_e,7} = -1, \quad c_{h_e,2} \rightarrow p_{h_e,2} = -10^{-4}, \quad c_{h_e,7} \rightarrow p_{h_e,1} = -1$$

$$-\frac{\sqrt{g}}{h_y} k_{ey} \frac{\partial T_e}{h_y \partial y} \Big|_{S_0} = \left[\left(p_{h_e,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{T_e}{m_e}} \right) T_e n_e \right] \Big|_{S_0}, \quad \text{bcene}=22$$

BCENE=9 (electron temperature decay length)

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \Big|_{S_0} = \begin{cases} \left[\frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,0}^{no_mdf} T_e - \frac{\sqrt{g}}{h_y} \frac{\kappa_e^{(AN)}}{p_{h_e,1}} T_e + \frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no_mdf} n_a T_e \right] \Big|_{S_0}, & \text{mdf_fhe} = 0 \\ \left[\frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no_mdf} T_e - \frac{\sqrt{g}}{h_y} \frac{\kappa_e^{(AN)}}{p_{h_e,1}} T_e + \frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no_mdf} n_a T_e - q_{ey}^{P.Sch} + \frac{\sqrt{g}}{h_y} \frac{3}{2} V_{ey}^{(E)} n_e T_e \right] \Big|_{S_0}, & \text{mdf_fhe} = 1 \end{cases}$$

South 1. Core

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \Big|_{S_1} = \left[\left(c_{h_e,0} \frac{\sqrt{g}}{h_y} + c_{h_e,1} \frac{\sqrt{g}}{h_y} T_e \right) \right] \Big|_{S_1}$$

$$c_{h_e,0} = 4 \cdot 10^{13}, \quad c_{h_e,a,1} = -10^{30},$$

total electron energy flux (8)

$$\frac{\sqrt{g}}{h_y} q_{ey} \Big|_{S_1} = p_{h_e,1} \frac{\frac{\sqrt{g}}{h_y}}{\sum_{core} \left(\frac{\sqrt{g}}{h_y} \right)} \Big|_{S_1}, \quad p_{h_e,1} = 8 \cdot 10^5, \quad \text{dpc}$$

North

$$-\frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \Big|_N = \left[\left(c_{h_e,2} \frac{\sqrt{g}}{h_y} \sqrt{\frac{T_e}{m_e}} \right) T_e n_e + c_{h_e,7} \max \left(0, \frac{\sqrt{g}}{h_y} \tilde{f}_{ey} \right) T_e + q_{ey}^{P.Sch} \right] \Big|_N, \quad \text{bcene}=11$$

$$c_{h_e,2} = -10^{-4}, \quad c_{h_e,7} = -1$$

$$\frac{\sqrt{g}}{h_y} k_{ey} \frac{\partial T_e}{h_y \partial y} \Big|_N = \left[\left(p_{h_e,1} \frac{\sqrt{g}}{h_y} \sqrt{\frac{T_e}{m_e}} \right) T_e n_e \right]_N, \quad \text{bcene}=22$$

$$\text{BCENE}=9 \text{ (electron temperature decay length)} \quad p_{h_e,1} = 10^{-2}$$

$$-\frac{\sqrt{g}}{h_y} \tilde{q}_{ey} \Big|_N = \begin{cases} \left[\frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,0}^{no_mdf} T_e - \frac{\sqrt{g}}{h_y} \frac{\kappa_e^{(AN)}}{p_{h_e,1}} T_e + \frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no_mdf} n_a T_e \right]_N, & \text{mdf_fhe} = 0 \\ \left[\frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,0}^{no_mdf} T_e - \frac{\sqrt{g}}{h_y} \frac{\kappa_e^{(AN)}}{p_{h_e,1}} T_e + \frac{5}{2} \sum_{a=0}^{n_s-1} z_a S_{n_a,1}^{no_mdf} n_a T_e + q_{ey}^{P.Sch} - \frac{\sqrt{g}}{h_y} \frac{3}{2} V_{ey}^{(E)} n_e T_e \right]_N, & \text{mdf_fhe} = 1 \end{cases}$$

West

bcene_15_style.eq.0 is connected with BCCON=14

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ex} \Big|_W = \left[- \left(\frac{1+\gamma}{1-\gamma} T_e + e\phi \right) \max \left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0, z_b \neq 0}^{n_s-1} m_b n_b}} n_a + \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_{||}}{e} \right) - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]_W \quad \text{bcene}=15$$

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ex} \Big|_W = \left[- \left(\frac{1+\gamma}{1-\gamma} + \frac{e\phi}{T_e} \right) \max \left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0, z_b \neq 0}^{n_s-1} m_b n_b}} n_a + \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_{||}}{e} \right) T_e - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]_W \quad \text{bcene}=15$$

bcene_15_style.eq.1

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ex} \Big|_W = \left[- \frac{\sqrt{g}}{h_x} |b_x| \left((1+\gamma) T_e + (1-\gamma) e\phi \right) n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp \left(- \frac{e\phi}{T_e} \right) - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]_W \quad \text{bcene}=15$$

sheath electron (3)

$$p_{m,a_{main},2} < 0.5, \text{ sound_mode } l=0,$$

if $p_{h_e,1} > 0$ then

$$\frac{\sqrt{g}}{h_x} q_{ex}|_W = \left[p_{h_e,1} + e \frac{\varphi - p_{\varphi,2}}{T_e} \right] \frac{\sqrt{g}}{h_x} v_{e_{out}} T_e - \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \Big|_W, \quad p_{h_e,1} = 0.9, \quad p_{\varphi,2} = 0,$$

$$v_{e_{out}} = \min \left\{ 0, -u_{u_{out}} n_e - \frac{j_x}{e} \right\}, \quad u_{u_{out}} = p_{m,a_{main},1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} m_b n_b}},$$

endif

East

bcene_15_style.eq.0 is connected with BCCON=14

$$-\frac{\sqrt{g}}{h_x} \tilde{q}_{ex}|_E = \left[-\left(\frac{1+\gamma}{1-\gamma} T_e + e\varphi \right) \max \left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} m_b n_b}} n_a - \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_{||}}{e} \right) + \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]_E \quad \text{bcene}=15$$

$$-\frac{\sqrt{g}}{h_x} \tilde{q}_{ex}|_E = \left[-\left(\frac{1+\gamma}{1-\gamma} + \frac{e\varphi}{T_e} \right) \max \left(0, \frac{\sqrt{g}}{h_x} \sum_{a=0}^{n_s-1} z_a p_{n,a,1} |b_x| \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} m_b n_b}} n_a - \frac{\sqrt{g}}{h_x} \frac{\tilde{j}_{||}}{e} \right) T_e + \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]_E \quad \text{bcene}=15$$

bcene_15_style.eq.1

$$-\frac{\sqrt{g}}{h_x} \tilde{q}_{ex}|_E = \left[-\frac{\sqrt{g}}{h_x} |b_x| ((1+\gamma)T_e + (1-\gamma)e\varphi) n_e \frac{1}{\sqrt{2\pi}} \sqrt{\frac{T_e}{m_e}} \exp\left(-\frac{e\varphi}{T_e}\right) + \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \right]_E \quad \text{bcene}=15$$

sheath electron (3)

$$p_{m,a_{main},2} < 0.5, \text{ sound_mode } l=0,$$

if $p_{h_e,1} > 0$ then

$$-\frac{\sqrt{g}}{h_x} \tilde{q}_{ex}|_E = - \left[p_{h_e,1} + e \frac{\varphi - p_{\varphi,2}}{T_e} \right] \frac{\sqrt{g}}{h_x} v_{e_{out}} T_e + \frac{\sqrt{g}}{h_x} q_{ex}^{P.Sch} \Big|_E, \quad p_{h_e,1} = 0.9, \quad p_{\varphi,2} = 0,$$

$$v_{e_{out}} = \max \left\{ 0, u_{out} n_e - \frac{j_x}{e} \right\}, \quad u_{out} = p_{m,a_{main},1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum_{b=0}^{n_s-1} p_b}{\sum_{b=0}^{n_s-1} m_b n_b}}, \quad z_b \neq 0$$

endif

v. Energy balance equation for ions

$$S_{h_i} = S_{h_i,0} + S_{h_i,1} T_i + S_{h_i,2} n_i + S_{h_i,3} n_i T_i$$

South. Private region

$$p_{h_i,1} = 10^{-2}, \quad e_{rec,0} = 0, \quad e_{rec,1} = 0.3, \quad r_{rec,0,0} = 0, \quad r_{rec,1,0} = 1$$

BCENE=9 (electron temperature decay length)

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{iy}|_{S_0} = \left[\begin{aligned} & \frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,0}^{no_mdf} T_i - \frac{\sqrt{g}}{h_y} \frac{\kappa_i^{(AN)}}{p_{h_i,1}} T_i + \frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,1}^{no_mdf} n_a T_i + \\ & + \sum_{a=0}^{n_s-1} \max \left\{ e_{rec,a} \left(\frac{3}{2} + B_{oRiS} \right) (z_{a,f} T_{e,f} + T_{i,f}), \left(\frac{3}{2} + B_{oRiS} \right) T_{Surf} k_B \right\} r_{rec,a,0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \end{aligned} \right]_{S_0}, \quad mdf_fhi = 0$$

$$\left[\begin{aligned} & \frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,0}^{no_mdf} T_i - \frac{\sqrt{g}}{h_y} \frac{\kappa_i^{(AN)}}{p_{h_i,1}} T_i + \frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,1}^{no_mdf} n_a T_i - q_{iy}^{P.Sch} + \frac{\sqrt{g}}{h_y} \frac{3}{2} \sum_{a=0}^{n_s-1} V_{ay}^{(E)} n_a T + \\ & + \sum_{a=0}^{n_s-1} \max \left\{ e_{rec,a} \left(\frac{3}{2} + B_{oRiS} \right) (z_{a,f} T_{e,f} + T_{i,f}), \left(\frac{3}{2} + B_{oRiS} \right) T_{Surf} k_B \right\} r_{rec,a,0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \end{aligned} \right]_{S_0}, \quad mdf_fhi = 1$$

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{iy}|_{S_0} = p_{h_i,2} \sum_{a=0}^{n_s-1} \left(\frac{\sqrt{g}}{h_y} n_a \sqrt{\frac{z_a T_e + T_i}{m_a}} \right) T_i \Big|_{S_0} + p_{h_i,1} \max \left(0, -\frac{\sqrt{g}}{h_y} \tilde{f}_{iy} \right) T_i \Big|_{S_0}, \quad bceni=21$$

$$p_{h_i,1} = -1, \quad p_{h_i,2} = -10^{-2}$$

$$-\frac{\sqrt{g}}{h_y} k_{iy} \frac{\partial T_i}{h_y \partial y} \Big|_{S_0} = p_{h_i,1} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \left(\frac{\sqrt{g}}{h_y} n_a \sqrt{\frac{z_a T_e + T_i}{m_a}} \right) T_i \Big|_{S_0} ,$$

bceni=22

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \Big|_{S_0} = \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \left(c_{h_i,a,2} \frac{\sqrt{g}}{h_y} \frac{n_a}{n_i} \sqrt{\frac{z_a T_e + T_i}{m_a}} \right) n_i T_i \Big|_{S_0} + c_{h_i,0,7} \max \left(0, -\frac{\sqrt{g}}{h_y} \tilde{f}_{iy} \right) T_i \Big|_{S_0} ,$$

b2stbc_spb

$$c_{h_i,a,2} = -10^{-2}$$

South 1. Core

total ion energy flux (8)

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \Big|_{S_1} = \left[p_{h_i,1} \frac{\frac{\sqrt{g}}{h_y}}{\sum_{core} \left(\frac{\sqrt{g}}{h_y} \right)} - q_{iy}^{P.Sch} + \frac{3}{2} \frac{\left(j_y^{(AN)} + j_y^{(in)} + \tilde{j}_y^{(vis||)} + \tilde{j}_y^{(visq)} + j_y^{(vis\perp)} \right)}{e} T_i \right] \Big|_{S_1} ,$$

$$\frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \Big|_{S_1} = \sum_{a=0}^{n_s-1} \left[\left(c_{h_i,a,0} \frac{\sqrt{g}}{h_y} \frac{n_a}{n_i} + c_{h_i,a,1} \frac{\sqrt{g}}{h_y} \frac{n_a}{n_i} T_i \right) \right] \Big|_{S_1}$$

$$c_{h_i,a,0} = 4 \cdot 10^{13}, c_{h_i,a,1} = -10^{30}, a = 0,1,$$

North

BCENE=9 (electron temperature decay length)

$$p_{h_i,1} = 10^{-2}, e_{rec,0} = 0, e_{rec,1} = 0.3, r_{rec,0,0} = 0, r_{rec,1,0} = 1$$

$$-\frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \Big|_N = \left\{ \begin{aligned} & \left[\frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,0}^{no_mdf} T_i - \frac{\sqrt{g}}{h_y} \frac{\kappa_i^{(AN)}}{p_{h_i,1}} T_i + \frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,1}^{no_mdf} n_a T_i + \right. \\ & \left. + \sum_{a=0}^{n_s-1} \max \left\{ e_{rec,a} \left(\frac{3}{2} + B_{oRis} \right) (z_{a,f} T_{e,f} + T_{i,f}) \left(\frac{3}{2} + B_{oRis} \right) T_{Surf} k_B \right\} r_{rec,a,0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \right]_N, \quad md_f - fhi = 0 \\ & \left[\frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,0}^{no_mdf} T_i - \frac{\sqrt{g}}{h_y} \frac{\kappa_i^{(AN)}}{p_{h_i,1}} T_i + \frac{5}{2} \sum_{a=0}^{n_s-1} S_{n_a,1}^{no_mdf} n_a T_i + q_{iy}^{P.Sch} - \frac{\sqrt{g}}{h_y} \frac{3}{2} \sum_{a=0}^{n_s-1} V_{ay}^{(E)} n_a T + \right. \\ & \left. + \sum_{a=0}^{n_s-1} \max \left\{ e_{rec,a} \left(\frac{3}{2} + B_{oRis} \right) (z_{a,f} T_{e,f} + T_{i,f}) \left(\frac{3}{2} + B_{oRis} \right) T_{Surf} k_B \right\} r_{rec,a,0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_y} \Gamma_{ay} \right\} \right]_N, \quad md_f - fhi = 1 \end{aligned} \right.$$

$$-\frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \Big|_N = p_{h_{i,2}} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \left(\frac{\sqrt{g}}{h_y} n_a \sqrt{\frac{z_a T_e + T_i}{m_a}} \right) T_i \Big|_N + p_{h_{i,1}} \max \left(0, -\frac{\sqrt{g}}{h_y} \tilde{f}_{iy} \right) T_i \Big|_N,$$

bceni=21

$$p_{h_{i,1}} = -1, \quad p_{h_{i,2}} = -10^{-2}$$

$$\frac{\sqrt{g}}{h_y} k_{iy} \frac{\partial T_i}{h_y \partial y} \Big|_N = p_{h_{i,1}} \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \left(\frac{\sqrt{g}}{h_y} n_a \sqrt{\frac{z_a T_e + T_i}{m_a}} \right) T_i \Big|_N, \quad \text{bceni}=22$$

$$-\frac{\sqrt{g}}{h_y} \tilde{q}_{iy} \Big|_N = \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} \left(c_{h_i,a,2} \frac{\sqrt{g}}{h_y} \frac{n_a}{n_i} \sqrt{\frac{z_a T_e + T_i}{m_a}} \right) n_i T_i \Big|_N + c_{h_i,0,7} \max \left(0, -\frac{\sqrt{g}}{h_y} \tilde{f}_{iy} \right) T_i \Big|_N, \quad \text{b2stbc_spb}$$

$$c_{h_i,a,2} = -10^{-2}$$

West

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ix} \Big|_W = -p_{h_{i,1}} \frac{\sqrt{g}}{h_x} |b_x| \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b} T_i \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} p_{n_a,1} n_a - \frac{\sqrt{g}}{h_x} q_{ix}^{P.Sch}} \Big|_W, \quad \text{bceni}=15$$

$$p_{h_i,1} = +1.5$$

sheath ion (3)

BCENI=3

$$p_{m,a_{main},2} = 0 < 0.5, \text{ sound_mode } l=0,$$

$$\frac{\sqrt{g}}{h_x} \tilde{q}_{ix}|_W = \left[\sum_{z_a \neq 0} \left(\frac{\sqrt{g}}{h_x} \frac{p_{h_i,2}}{2} u_{u_{out}} m_a n_a \left(u_{p_{out}}^2 + \left(u_{a,x}^{(dia)} \right)^2 \right)_i - \frac{\sqrt{g}}{h_x} p_{h_i,1} u_{u_{out}} n_a T_i \right) + \sum_{a=0}^{n_s-1} \max \left\{ c_{rec,a,1} \left(\frac{3}{2} + B_{oRiS} \right) (z_{a,f} T_{e,f} + T_{i,f}), \left(\frac{3}{2} + B_{oRiS} \right) T_{Surf} k_B \right\} r_{rec,a,0} c_{nsr} \max \left\{ 0, -\frac{\sqrt{g}}{h_x} \Gamma_{ax} \right\} \right]_W$$

$$u_{a,x}^{(dia)} = V_{ax}^{(E)} + \tilde{V}_{xa}^{(dia)}$$

$$u_{p_{out}} = \max \left\{ 0, \frac{u_{u_{out}}}{|b_x|} + \frac{|B_z|}{|B_x|} u_{1,x}^{(dia)} \right\}$$

$$u_{u_{out}} = p_{m,a_{main},1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}}, \quad p_{m,a_{main},1} = 1.0$$

if neutrals_namelist.eq.1 .or. use_eirene.ne.0 then $e_{rec,0} = 0$, $e_{rec,1} = 0.3$, $r_{rec,0} = 0$, $r_{rec,1} = 1$ **else** $c_{rec,0,0} = 0$,

$$c_{rec,1,0} = 1, \quad c_{rec,0,1} = 0, \quad c_{rec,1,1} = 0.3$$

$$p_{h_i,1} = 2.5 \text{ but it should be } p_{h_i,1} = 1.5$$

$$p_{h_i,2} = 0 \text{ without drifts}$$

$$p_{h_i,2} = 10^{-3} \text{ with drifts but it should be } p_{h_i,2} = 0$$

East

$$-\frac{\sqrt{g}}{h_x} \tilde{q}_{ix}|_E = -p_{h_i,1} \frac{\sqrt{g}}{h_x} |b_x| \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}} T_i \sum_{\substack{a=0 \\ z_a \neq 0}}^{n_s-1} p_{n,a,1} n_a - \frac{\sqrt{g}}{h_x} q_{ix}^{P.Sch} \quad \text{BCENI=15}$$

$$p_{h_i,1} = +1.5$$

sheath ion (3)

BCENI=3

$$p_{m,a_{main},2} = 0 < 0.5, \text{ sound_mode } l=0,$$

$$-\frac{\sqrt{g}}{h_x} \tilde{q}_{ix}|_E = \left[\sum_{z_a \neq 0} \left(-\frac{\sqrt{g}}{h_x} \frac{p_{h_i,2}}{2} u_{out} m_a n_a \left(u_{p_{out}}^2 + (u_{a,x}^{(dia)})^2 \right) - \frac{\sqrt{g}}{h_x} p_{h_i,1} u_{out} n_a T_i \right) + \sum_{a=0}^{n_s-1} \max \left\{ c_{rec,a,1} \left(\frac{3}{2} + B_{oRis} \right) (z_{a,f} T_{e,f} + T_{i,f}), \left(\frac{3}{2} + B_{oRis} \right) T_{Surf} k_B \right\} c_{rec,a,0} c_{nsr} \max \left\{ 0, \frac{\sqrt{g}}{h_x} \Gamma_{ax} \right\} \right]_E$$

$$u_{a,x}^{(dia)} = V_{ax}^{(E)} + \tilde{V}_{xa}^{(dia)}$$

$$u_{p_{out}} = \max \left\{ 0, \frac{u_{out}}{|b_x|} - \frac{|B_z|}{|B_x|} u_{1,x}^{(dia)} \right\}$$

$$u_{out} = p_{m,a_{main},1} c_s |b_x|, \quad c_s = \sqrt{\frac{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} p_b}{\sum_{\substack{b=0 \\ z_b \neq 0}}^{n_s-1} m_b n_b}}, \quad p_{m,a_{main},1} = 1.0$$

if neutrals_namelist.eq.1 .or. use_eirene.ne.0 then $e_{rec,0} = 0$, $e_{rec,1} = 0.3$, $r_{rec,0} = 0$, $r_{rec,1} = 1$ **else** $c_{rec,0,0} = 0$,

$c_{rec,1,0} = 1$, $c_{rec,0,1} = 0$, $c_{rec,1,1} = 0.3$

$p_{h_i,1} = 2.5$ **but it should be** $p_{h_i,1} = 1.5$

$p_{h_i,2} = 0$ **without drifts**

$p_{h_i,2} = 10^{-3}$ **with drifts but it should be** $p_{h_i,2} = 0$

Appendix

Old form of $(\nabla \pi_a^\parallel)_\parallel$ was

$$(\nabla \pi_a^\parallel)_\parallel = \begin{cases} 0, & a \neq 1 \\ -\frac{4}{3} b_x B^{\frac{3}{2}} \frac{\partial}{h_x \partial x} \left(\frac{c_{\eta_{ax}^{(CL)}} c_{\eta_{ax}^{(CL)}}^{(Luciani)} \eta_{CLax} b_x}{B^2} \frac{\partial \left(B^{\frac{1}{2}} \left(V_{a\parallel} + 0 \cdot \frac{B}{B_x} V_{ax}^{(dia)} + 0 \cdot \frac{B}{B_x} V_x^{(E)} \right) \right)}{h_x \partial x} \right) - \\ - 0 \cdot B^{\frac{3}{2}} b_x \frac{\partial}{h_x \partial x} \left(\frac{b_x}{v_{aa} B^2} \frac{\partial \left(B^{\frac{1}{2}} \left(q_{a\parallel}^{(0)} + \frac{B}{B_x} q_{ax}^{(dia)} \right) \right)}{h_x \partial x} \right) \end{cases}, \quad a = 1$$

Contribution from anomalous viscosity was contained in l.h.s.

Because of $\eta_{ANa} \ll \tilde{\eta}_{CLax}$, $\eta_{ax} \approx \tilde{\eta}_{CLax}$ and taking into account $B_x h_x h_y = const$

$$-\frac{4}{3} b_x B^{\frac{3}{2}} \frac{\partial}{h_x \partial x} \left(\frac{c_{\eta_{ax}^{(CL)}} c_{\eta_{ax}^{(CL)}}^{(Luciani)} \eta_{CLax} b_x}{B^2} \frac{\partial \left(B^{\frac{1}{2}} V_{a\parallel} \right)}{h_x \partial x} \right) = \frac{\partial}{h_z \sqrt{g} \partial x} \left(\frac{h_z \sqrt{g}}{h_x} \left(\frac{4}{3} \eta_{ax} \frac{\partial \ln h_z}{h_x \partial x} V_{\parallel a} - \frac{4}{3} \eta_{ax} \frac{\partial V_{\parallel a}}{h_x \partial x} \right) \right) -$$

$$-\frac{\partial}{h_z \sqrt{g} \partial x} \left(\frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \eta_{ax} \frac{\partial \ln \left(h_z B^{\frac{1}{2}} \right)}{B^{\frac{1}{2}} h_x \partial x} \right) B^{\frac{1}{2}} V_{\parallel a}$$

New form of $(\nabla \pi_a^\parallel)_\parallel$

$$(\nabla \pi_a^\parallel)_\parallel = \begin{cases} 0, & a \neq 1, \\ \frac{\partial}{h_z \sqrt{g} \partial x} \left(\frac{h_z \sqrt{g}}{h_x} \left(\frac{4}{3} \eta_{ax} \frac{\partial \ln h_z}{h_x \partial x} V_{\parallel a} - \frac{4}{3} \eta_{ax} \frac{\partial V_{\parallel a}}{h_x \partial x} \right) \right) - \frac{\partial}{h_z \sqrt{g} \partial x} \left(\frac{h_z \sqrt{g}}{h_x} \frac{4}{3} \eta_{ax} \frac{\partial \ln \left(h_z B^{\frac{1}{2}} \right)}{B^{\frac{1}{2}} h_x \partial x} \right) B^{\frac{1}{2}} V_{\parallel a} \end{cases}, \quad a = 1$$

About scaling of residuals

$$\begin{aligned}
& \frac{\sum_{i=0}^{n_x-1} \sum_{j=0}^{n_y-1} |r_{n_{a,i,j}}|}{\sum_{i=0}^{n_x-1} \sum_{j=0}^{n_{xy}-1} \sqrt{g_{i,j}} n_{a,i,j}}, \\
& \frac{\sum_{i=0}^{n_x-1} \sum_{j=0}^{n_y-1} |r_{u_{a,i,j}}|}{\sum_{i=0}^{n_x-1} \sum_{j=0}^{n_{xy}-1} \sqrt{g_{i,j}} n_{a,i,j} \sqrt{m_a T_i}}, \\
& \frac{\sum_{i=0}^{n_x-1} \sum_{j=0}^{n_y-1} |r_{h_{e,i,j}}|}{\frac{3}{2} \sum_{a=0}^{n_s-1} \sum_{i=0}^{n_x-1} \sum_{j=0}^{n_{xy}-1} \sqrt{g_{i,j}} n_{a,i,j} (z_a T_e + T_i)}, \\
& \frac{\sum_{i=0}^{n_x-1} \sum_{j=0}^{n_y-1} |r_{h_{i,i,j}}|}{\frac{3}{2} \sum_{a=0}^{n_s-1} \sum_{i=0}^{n_x-1} \sum_{j=0}^{n_{xy}-1} \sqrt{g_{i,j}} n_{a,i,j} (z_a T_e + T_i)}, \\
& \frac{\sum_{i=0}^{n_x-1} \sum_{j=0}^{n_y-1} |r_{po_{i,j}}|}{\sum_{a=0}^{n_s-1} \sum_{i=0}^{n_x-1} \sum_{j=0}^{n_{xy}-1} \sqrt{g_{i,j}} z_a e n_{a,i,j}}
\end{aligned}$$